Atomic Visibility Is the Price of Mutable Redundancy: Schema, Placement and Write Order on Sharded SQL Engines

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ABSTRACT

Sharded SQL engines such as Citus and Vitess, like applications that split their data across databases, keep no global snapshot. Hence plain SQL readers may see intermediate states of a distributed update that violate the constraints, even under two-phase commit. Atomic visibility hides these states but costs global coordination or a rewrite of every reader, whereas an order of the writes needs neither. We initiate research on order-sufficiency: given the attributes that a workload keeps immutable, every modification has a legal write order. Our central result locates the need for atomic visibility in mutable redundancy. Under functional dependencies and foreign keys with cached columns, an update that writes fresh values needs it exactly when it rewrites a value and its copy. For arbitrary modifications, order-sufficiency is undecidable, but decidable in polynomial time for functional dependencies and three classes of inclusion dependencies. Choosing the immutable attributes is NP-hard but a minimum cut when foreign keys with cached columns form a forest, and these attributes decide the admissible shard keys. Write orders compile to leveled commits, which use the fewest rounds and, sent straight to the shards, take less time than two-phase commit. In GitLab, Sentry and ten further applications, most multi-table write templates need only order, and almost all others copy or derive mutable attributes. On Vitess, and on Citus once one worker commits a millisecond late, one two-phase commit exposes orphans, which leveled commits hide from readers that read first what an update writes last or finish within a pause between commits.

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The source code, data, and/or other artifacts have been made available at https://github.com/zhuoxingzhang/order-sufficiency.

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1 INTRODUCTION

Sharded SQL engines such as Citus [24] and Vitess [74] scale relational applications by partitioning every table into shards. A transaction that writes several shards commits by two-phase commit (2PC), but each shard makes its writes visible at its own commit point. No global snapshot hides the difference, so 2PC makes such a transaction atomic but not atomically visible. Plain SQL readers observe its intermediate states. On Vitess, for example, a reader that saw a new child found no parent in 86% of the runs whose 2PC wrote the child first (Sec. 6.3). Applications that split their data across databases, such as GitLab and Sentry, expose the same states.

Whether these states harm an application depends on the constraints that its schema promises. The engine checks keys and foreign keys within a shard, but the application also relies on foreign keys across shards, on functional dependencies (FDs) and on copies of attributes. States that violate them do harm in practice. GitLab leaves the children of a deleted parent in its other databases to a background task. A migration that read such a dangling child, an intermediate state of the deletion, aborted an upgrade in a severity-one incident [34]. An update whose intermediate states may violate these constraints needs atomic visibility, unless its writes can be ordered so that every state that readers see is legal.

Providing atomic visibility is costly. Engines with global snapshots, such as Spanner [22] and CockroachDB [68], provide it, but every distributed transaction then pays for global coordination [13]. RAMP transactions [6, 7], as deployed in RAMP-TAO [20], add it on top of 2PC, but they rewrite every reader and attach metadata to every read. Highly available transactions [3] and causal consistency [8, 47, 48] likewise rewrite every reader, and their metadata is at best of constant size [49]. Indeed, reads cannot be atomically visible, minimal-delay and fresh at once [70]. We keep the readers as they are: plain SQL statements of object-relational mappers, reports and other services. The writer must then keep the states that they see legal by the order of its writes alone. Whether such an order exists depends on the update, as the following example shows.

Example 1.1 (Groups and projects). Relation $Group(gid, path)$ stores groups with their URL path, and relation $Project(pid, gid, gpath)$ stores projects with their group and a cached copy of the group's path, much like the routes of GitLab [35] (Sec. 6.2). The constraints are a foreign key and an inclusion dependency (IND) that keeps the cached copy correct:

$$Project[gid] \subseteq Group[gid],$$
$$Project[gid, gpath] \subseteq Group[gid, path].$$

Groups are sharded by *gid* and projects by *pid*, and the workload creates groups and projects, moves projects between groups, and

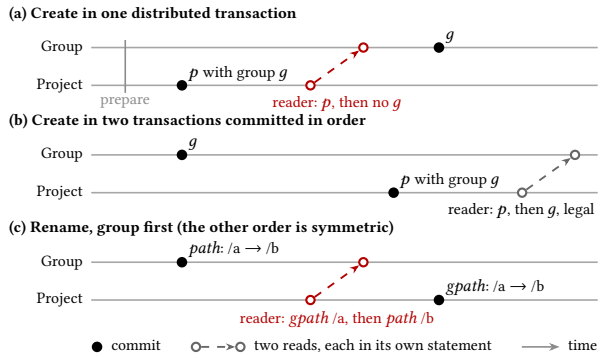


Figure 1: Commits and Reads for the Updates of Ex. 1.1

renames groups. Creating a group with its first project writes two shards. If both writes commit in one distributed transaction, a reader may see the project before its group. If the application commits the group first, a reader that reads the project and then its group sees no orphan. Hence, for creation, the order of the writes and not their atomicity decides which states readers see. Renaming a group changes the group and every one of its projects. Whichever is written first, a reader may see a project whose cached path matches no group, so no order helps.

The example yields three observations, which Fig. 1 illustrates on two shards. Firstly, for readers that read a project before its group, write order prevents an anomaly that atomic commit does not. For insertions and deletions, the classical rule of writing referenced entities first and deleting them last, which object-relational mappers apply, gives such an order. Secondly, some modifications admit no legal order, and no rule tells which. This depends on the schema and the workload: without the cached copy, or if groups were never renamed, every modification of Ex. 1.1 would be harmless in any order. Thirdly, placement can substitute for atomic visibility: if projects never move and are sharded by their group, a rename touches one shard and commits atomically. This raises two questions. (Q1) Which updates of a workload need atomic visibility because no write order keeps the constraints satisfied, and can this be decided? (Q2) How should a designer choose the attributes that the workload modifies, the shard keys and the write orders, so that fewer updates need atomic visibility and the others stay legal after every commit?

Previous work does not answer these questions. Invariant confluence [4] decides whether concurrent operations need coordination, not whether the writes of one update can be ordered so that readers see only legal states. Transaction chopping [65] and transaction chains [80] commit the pieces of a transaction in order, but they analyze all transactions, readers included, whereas the readers of a sharded engine are arbitrary SQL statements. We are unaware of work that characterizes when write order alone keeps the constraints of a schema satisfied.

We treat order-sufficiency, the existence of a legal write order for every modification, as a property of the schema. It is relative to the attributes that a workload never modifies, which we call frozen. Our central result locates the need for atomic visibility in mutable redundancy. Under FDs and foreign keys with cached columns, an update that writes fresh values, as renames and new tokens do, needs atomic visibility exactly when it rewrites a value and its copy (Thm. 4.7). If modifications may also exchange values, such a schema either makes every modification legal in any order or admits one that needs atomic visibility, so write order adds nothing there. Classical normal forms emerge as special cases. Under FDs, every fresh update is legal in any order exactly on schemas in Boyce-Codd normal form, and every modification outside the primary keys exactly on schemas in the object normal form of Biskup [14]. Order matters instead for insertions, deletions and their mixtures, such as re-keying a group by inserting a new group, moving its projects and deleting the old one. It also matters for one-way conditions on a status, such as an archived flag. Hence a design proceeds in three steps. Freezing an attribute replaces its modifications by re-keying, shard keys keep the modifications that still need atomic visibility within one shard, and leveled commits write every other update in a legal order. They commit single-shard transactions in rounds, adding a round wherever the order crosses shards.

Real applications follow this pattern. We call a code location that writes across databases a template (Sec. 2.2). In GitLab [35] and Sentry [64], 28 of 32 and 48 of 65 such templates need only order. Every template that needs atomic visibility maintains a copy or a derived value of a mutable attribute, and our static analyzer finds the same pattern in ten further applications. However, write order protects only readers that read first what an update writes last. It alone protects every reader of 47 of the 76 order-only templates. In the other 29, a reader reads the other way, and a pause between commits protects it only if it finishes within the pause.

We summarize our contributions as follows.

(1) Conceptually, we introduce order-sufficiency, that some write order keeps every state legal, and order-insensitivity, that every order does, relative to the frozen attributes. Implication by update-stable consequences, those that no modification can violate, suffices for all FDs and INDs, and it is exact for FDs (Secs. 2 and 3).
(2) Theoretically, we prove that a fresh update needs atomic visibility exactly when it rewrites a value and its copy, under FDs and under a class of acyclic foreign keys with cached columns. For arbitrary modifications, order-sufficiency is undecidable for INDs but decidable in polynomial time for unary INDs, for acyclic INDs that reference keys and for this class. On these classes, as for FDs, it coincides with order-insensitivity. Conditional INDs, which keep one-way conditions on a status, give every update a legal order (Sec. 4).
(3) Algorithmically, we show that choosing the attributes to freeze is NP-hard but a minimum cut when foreign keys with cached columns form a forest. Shard keys preserve executability exactly when the frozen attributes of every FD that modifications can violate determine them, and every reference that needs atomic visibility stores the shard key of the entity that it references. Write orders compile to leveled commits, which need the fewest rounds and protect every reader that finishes within a pause between rounds. After a crash, they complete or stop the update in a legal state without overwriting others (Sec. 5).
(4) Experimentally, we find that one 2PC exposes orphans on Vitess, and on Citus once one worker commits 1 ms late, whereas

leveled commits expose none to the readers that write order protects. Sent straight to the shards, leveled commits also take less time than 2PC for up to six rounds. On GitLab, co-location on the declared sharding keys carries every reference that needs atomic visibility as long as these keys stay frozen. Optimal freezing saves more the more skewed the frequencies of updates are (Sec. 6).

Overall, for fresh updates atomic visibility is the price of mutable redundancy, which freezing and placement can remove, while write order keeps every other update legal after each commit.

Organization. Sec. 2 fixes the engine model and defines order-sufficiency. Secs. 3 and 4 answer (Q1) for FDs and INDs, with the central result in Sec. 4.2, and Sec. 5 answers (Q2) by freezing attributes, choosing shard keys and compiling write orders. Sec. 6 tests these answers on GitLab, Sentry and ten further applications and on Citus and Vitess. Sec. 7 discusses related work, and Sec. 8 concludes. *The proofs and further results are in the full version,* ***order-sufficiency-full.pdf*** *in our artifact repository.*

2 MODEL AND PROBLEM

We fix the engine model under which all later statements hold, and then define order-sufficiency.

2.1 Sharded SQL Engines

A sharded SQL engine partitions every table into shards by a shard key and stores each shard on one node. In 2PC, all shards prepare, and each shard then makes the writes visible at its own commit point. The engine guarantees no order among these commit points and keeps no global snapshot. Citus [24] sends the second phase to all shards in parallel, and Vitess [74], in its 2PC mode [75], commits the shard that it touched first before the others. Hence both engines expose partial states (Sec. 6.3).

All results of this paper hold under five assumptions.

(1) *Readers* are plain SQL statements that are not rewritten. Each statement reads, on each shard, the latest state that the shard has committed, and it sees no version metadata. These are the minimal-delay readers of Tomsic et al. [70].
(2) *Storage* holds the logical relations, or projections of them, as physical tables, with one version of every entity and no version columns or commit flags.
(3) *Constraints* declared as primary keys, unique constraints and foreign keys within a shard are checked by the engine. Other constraints, such as foreign keys across shards and functional dependencies, are relied upon by readers but not checked.
(4) *Isolation* is native: every shard runs multiversion concurrency control, statements read only committed data, and nothing repairs reads.
(5) *Read order*: a reader that write order protects reads the changed entities of an update in the reverse of the order in which they are written, each in a later statement. That is, it reads first what the update writes last. Other readers are protected by updates whose every intermediate state is legal (Sec. 2.3), or by pauses between commits that outlast their reads (Sec. 5.3).

Both engines meet assumptions 1 and 4 by default. Citus runs at read committed, and Vitess runs MySQL at repeatable read on every shard, where a statement outside a transaction reads the latest committed state [74]. Whether a read path meets assumption 5 depends on its direction. Following a foreign key from a child to its parent reverses an insertion, and listing the children of a parent reverses a deletion. No write order protects the same reads against the other update. In the applications of Sec. 6, which allow no joins across databases, readers differ in this direction alone.

2.2 Schemas, Workloads and Modification Updates

FDs and INDs have their standard meaning [1]. A schema consists of relations, each with a primary key $PK(R)$, and a set Σ of FDs and INDs. INDs may be composite and cyclic, and instances are finite. An *entity* is a relation together with a value of its primary key. We call the attributes that a workload never modifies *frozen*, and write $Fz(R)$ for the frozen attributes of a relation R, which always include $PK(R)$. The other attributes are *mutable*, and a choice of Fz is a *freeze*. Where no confusion arises, Fz also denotes the set of all frozen attributes, qualified by their relations, and similarly for primary and shard keys.

An update of the workload changes several entities, and a reader may see any mix of their old and new tuples. This motivates the following definition.

Definition 2.1 (Modification update). A *modification update* is a pair (I, J) of instances that both satisfy Σ, contain the same entities, and agree on the frozen attributes of every entity. Its *changed entities* Ch are those whose tuples differ in I and J. For $T \subseteq Ch$, the *intermediate state* $I \oplus T$ takes the tuples of J for the entities in T and the tuples of I for all other entities.

The intermediate state $I \oplus T$ is what a reader sees when the writes to the entities in T are visible and the others are not. For example, renaming a group of Ex. 1.1 with two projects changes three entities. One of its intermediate states shows the new path of the group next to the old cached paths of the projects. A *template* names the relations into which a code location inserts, those from which it deletes and the attributes that it modifies. Its updates are those that change entities accordingly, and it *needs only order*, or is order-only, if all its updates are executable in the sense defined next. Since a modification update is any pair of Def. 2.1, a workload enters our results only through Fz, and its modification updates include combinations of changes that no single template issues. Secs. 3 and 4 study modification updates, since updates that only insert or only delete follow the classical rule (Prop. 5.4). Thm. 4.8 and Sec. 5.3 extend the results to updates that mix them.

2.3 Order-Sufficiency

An application can restrict the mixes that readers see by writing the changed entities one at a time, committing each write before it issues the next. The storage then passes only through the prefix states $I \oplus \{e_1, \ldots, e_k\}$ of this order. Two notions arise: some order keeps every prefix state legal, or every intermediate state is legal.

Definition 2.2 (Order-sufficiency). A modification update is *executable* if some permutation $e_1, \ldots, e_n$ of Ch makes every prefix state $I \oplus \{e_1, \ldots, e_k\}$ satisfy Σ, and it is *order-insensitive* if every intermediate state $I \oplus T$ satisfies Σ. The schema Σ is *order-sufficient* under

Fz if every modification update is executable, and *order-insensitive* under *Fz* if every modification update is order-insensitive.

Every prefix state is an intermediate state, so order-insensitivity implies order-sufficiency. An order writes every entity once, since readers would also see any temporary value. We fix the language of orders as follows.

Definition 2.3 (Legal orders). An instance is *legal* if it satisfies Σ, and a permutation of *Ch* is a *legal order* if all its prefix states are legal. A modification update *needs atomic visibility* if it is not executable. Two changed entities *must become visible together* if every intermediate state that contains the new tuple of exactly one of them is illegal. For an IND $R[X] \subseteq S[Y]$, a *witness* of a tuple t of R is a tuple s of S with $s[Y] = t[X]$. An update is *binary* if some directed graph $\mathcal{P}$ on *Ch*, its *precedence graph*, makes an intermediate state $I \oplus T$ legal exactly when no edge of $\mathcal{P}$ leads from an entity outside T to an entity in T.

That is, an edge from v to u forbids every state in which u is new and v is old. An update is binary whenever every IND $R[X] \subseteq S[Y]$ in Σ has a key K of S inside Y and the update hands no value of K from one entity to another (Prop. A.1 in Appendix A.1). Foreign keys to primary keys are the typical case.

Order-sufficiency and order-insensitivity differ in which readers they protect. A reader that follows assumption 5, reading the changed entities in the reverse of the write order, observes a prefix state. Indeed, if it sees the new version of e_j, then every e_i with $i < j$ was committed before e_j and is read after it, so the reader also sees the new version of e_i. A reader that cannot choose its read order, such as a single join over several shards, can observe any intermediate state. Hence order-sufficiency is what write order supplies to the readers of assumption 5, and order-insensitivity is what an arbitrary reader needs. Neither notion is an isolation level. Both promise a reader only that every state it observes satisfies Σ, neither a snapshot nor serializability, and Sec. 5.3 extends the promise to readers that finish within a pause between commits. Which updates they cover depends on the update class: Secs. 3 and 4 decide them for arbitrary modifications, for fresh updates and for mixtures with insertions and deletions. In Ex. 1.1, every move is executable, whereas a rename of a group with projects is not, since the group and its projects must become visible together (Ex. A.2 in Appendix A.1).

Order-sufficiency thus depends on the schema and on the attributes that the workload modifies. The decision problem asks, given Σ and *Fz*, whether Σ is order-sufficient under *Fz*. The design problems ask which attributes to freeze and which shard keys to choose so that the schema becomes order-sufficient, or every update executable on its shards (Sec. 5).

3 CHARACTERIZING ORDER-SUFFICIENCY

Order-sufficiency is governed by the constraints that no modification of the workload can violate. This section identifies these constraints, shows that they suffice for every schema, and shows that they are exact for FDs. Throughout, implication means finite implication: a set Γ of constraints implies Σ, written $\Gamma \models \Sigma$, if every finite instance that satisfies Γ satisfies Σ. The closure Σ^+ is the set of FDs and INDs that Σ implies. For attributes X of a relation R, the closure X^+ is the set of attributes A of R with $\Sigma \models X \to A$.

3.1 Update-Stable Consequences

Freezing an attribute protects more than the attribute itself. Intuitively, consider an IND that copies attributes from a referenced entity that frozen values identify. The referenced entity is then the same before and after the update, and so are the copies of its frozen attributes. This motivates the following closure.

Definition 3.1 (Frozen closure). The *frozen closure* M^* is the least family of attribute sets such that $M^*(R) \supseteq Fz(R)$ for every relation R and that is closed under the following rule for every IND $R[X] \subseteq S[Y]$ in Σ^+. If Y contains a key K of S with $K \subseteq M^*(S)$, and the attributes of X at the positions of K belong to $M^*(R)$, then $X_i \in M^*(R)$ for every position i with $Y_i \in M^*(S)$.

That is, the closure adds every attribute that copies a frozen attribute of an entity that frozen values identify. Its significance lies in the following invariant.

Lemma 3.2 (Invariance of the frozen closure). *Every modification update leaves the attributes in M^* of every entity unchanged.*

Constraints that involve only such unchanged values, or frozen keys, survive every modification update.

Definition 3.3 (Update-stable consequences). The set $\mathcal{G}$ of *update-stable consequences* contains (i) every FD $X \to A$ in Σ^+ such that X contains a key inside M^* or $X \cup \{A\} \subseteq M^*$, and (ii) every IND $R[X] \subseteq S[Y]$ in Σ^+ such that $Y \subseteq M^*(S)$, or such that $X \subseteq M^*(R)$ and Y contains a key of S inside $M^*(S)$.

Each constraint in $\mathcal{G}$ holds in every intermediate state of every modification update. Indeed, a violation would involve only values that the update does not change, by Lem. 3.2, or two tuples that would already violate a frozen key in I. In Ex. 1.1, $\mathcal{G}$ contains the foreign key but not the cached-path IND, exactly the constraint that a rename violates (Ex. A.3 in Appendix A.2).

3.2 Sufficiency

Update-stable consequences suffice for order-insensitivity, whatever the FDs and INDs.

Theorem 3.4 (Sufficiency). *Let Σ be any set of FDs and INDs, cyclic INDs included. If $\mathcal{G} \models \Sigma$, then Σ is order-insensitive, and hence order-sufficient, under Fz.*

Thm. 3.4 reduces a question about all updates of a workload to an implication problem over the schema. The converse fails for some schemas with INDs (Sec. 4.3) but holds for FDs.

3.3 Functional Dependencies

For FDs, the condition of Thm. 3.4 is also necessary and takes a syntactic form. Following Zaniolo [79], an FD $X \to A$ is *elementary* if $A \notin X$ and no proper subset of X determines A.

Theorem 3.5 (Exactness for FDs). *Let Σ contain only FDs. The following are equivalent: (1) Σ is order-sufficient under Fz; (2) Σ is order-insensitive under Fz; (3) $\mathcal{G} \models \Sigma$; (4) every elementary FD $X \to A$ in Σ^+ satisfies $X \subseteq Fz$, and also $A \in Fz$ if X is not a superkey. Condition (3) is decidable in polynomial time.*

The proof in Appendix A.2 yields more than the equivalence. With FDs only, either every modification update is order-insensitive, or some update has two changed entities that must become visible together. We illustrate it on a variant of our running example.

Example 3.6 (Paths stored in projects). Suppose the path is stored only in *Project*, under the FD $gid \to gpath$, and consider *Project* on its own. Condition (4) fails because *gid* is not a superkey and *gpath* is mutable. Indeed, the two projects of a renamed group must become visible together: whichever project is written first, the intermediate state separates the two projects on *gpath*. No order helps, and this design needs atomic visibility for every rename.

Condition (4) relates to normal forms, but it depends on the workload through *Fz*. For a schema in Boyce-Codd normal form (BCNF), it holds exactly when the workload freezes every minimal key. With only the primary key frozen, condition (4) forces the primary key to be the only minimal key and coincides with the object normal form of Biskup [14].

The updates that prove the necessity part of Thm. 3.5 exchange values between two entities whenever a key or a left side is mutable. Applications rarely do so, since renames, new email addresses and rotated tokens write values that no other row holds. We call a modification update *fresh* if every value that it writes into an attribute of the left side of an FD or of the right side of an IND in Σ is new in its column. Fresh updates relax condition (4) as follows.

Proposition 3.7 (Fresh updates). *Let Σ contain only FDs. Every fresh modification update is order-insensitive under Fz if and only if every elementary FD $X \to A$ in Σ^+ whose left side X is not a superkey has $A \in Fz$. Otherwise, some fresh update with two changed entities is not executable.*

Hence fresh updates need atomic visibility exactly where the schema stores a mutable attribute redundantly. In particular, every schema in BCNF is order-insensitive for fresh updates under every freeze. However, the design of Ex. 3.6 still needs atomic visibility, since a rename writes its new path into every project of the group. Thm. 4.7 in Sec. 4.2 extends this result to foreign keys with cached columns.

4 INCLUSION DEPENDENCIES

With INDs, the condition of Thm. 3.4 remains sufficient but is no longer necessary. Order-sufficiency even becomes undecidable, since finite implication of an FD by FDs and INDs is undecidable [18, 54], [1, Thm. 9.2.4]. Indeed, implication reduces to order-sufficiency and to order-insensitivity already when the workload modifies a single attribute (Prop. A.6 in Appendix A.3). The difficulty lies in implication, not in write order. Hence a decidable test must restrict the INDs. Sec. 4.1 identifies three classes on which the condition is exact and decidable in polynomial time. Sec. 4.2 characterizes fresh updates on the third, and Sec. 4.3 shows that conditional INDs give every update a legal order. Fig. 8 in Appendix A.3 summarizes the results by class.

4.1 Exact Classes

These classes share one conclusion with FDs. For arbitrary modifications, the condition of Thm. 3.4 is exact and decidable in polynomial time, and order-sufficiency coincides with order-insensitivity. They differ in how the INDs interact with the FDs. The *IND graph* of Σ links R to S for every IND $R[X] \subseteq S[Y]$ in Σ.

Unary INDs. Following Cosmadakis et al. [23], an IND $R[A] \subseteq S[B]$ with single attributes A and B is *unary*. Unary INDs and FDs interact through cycles, which can make an attribute a key in every finite instance (Ex. A.8). Armstrong's axioms, transitivity of unary INDs and such cycle rules axiomatize finite implication, which is decidable in polynomial time [23], [1, Thm. 9.4.7]. The following theorem shows that finite implication decides order-sufficiency for unary INDs, and that the write order never makes a difference.

Theorem 4.1 (Exactness for unary INDs). *Let Σ consist of FDs with nonempty left sides and unary INDs. The following are equivalent: (1) Σ is order-sufficient under Fz; (2) Σ is order-insensitive under Fz; (3) $\mathcal{G} \models \Sigma$; (4) every nontrivial unary IND in Σ^+ has a frozen right side, and every elementary FD $X \to A$ in Σ^+ satisfies $X \subseteq Fz$, and also $A \in Fz$ if X is not a superkey; (5) no modification update has two changed entities that must become visible together. Condition (4) is decidable in polynomial time.*

Key-referencing INDs. The second class captures foreign keys that reference keys. Let $\mathcal{A}$ contain the schemas whose IND graph is acyclic and whose INDs reference keys. That is, for every IND $R[X] \subseteq S[Y]$, the attributes Y form a minimal key of S under the FDs declared on S, and no relation is referenced through two different keys. In particular, acyclic foreign keys to primary keys yield schemas in $\mathcal{A}$. For these schemas, the FDs of each relation alone decide order-sufficiency.

Theorem 4.2 (Exactness for key-referencing INDs). *Let Σ belong to $\mathcal{A}$. The following are equivalent: (1) Σ is order-sufficient under Fz; (2) Σ is order-insensitive under Fz; (3) $\mathcal{G} \models \Sigma$; (4) for every relation R, the set F_R of FDs declared on R satisfies condition (4) of Thm. 3.5 in place of Σ; (5) every modification update with exactly two changed entities is executable. Condition (4) is decidable in polynomial time by Thm. 3.5.*

In Ex. 1.1, it is the cached copy, not the foreign key, that defeats write order (Ex. A.10 in Appendix A.3).

Foreign keys with cached columns. The third class adds what $\mathcal{A}$ excludes, namely copies of referenced attributes. An IND $R[PD] \subseteq S[LC]$ follows a pointer P to the primary key L of S and keeps every copy D_i equal to its source C_i.

Definition 4.3 (Class $\mathcal{K}$). A schema belongs to the class $\mathcal{K}$ if its only FDs are the primary keys, its IND graph is acyclic, and every IND has the form $R[PD] \subseteq S[LC]$ with $L = PK(S)$, where (i) no other IND has the *pointer* P and the referenced relation S, (ii) every *copy* D_i lies outside $PK(R)$ and occurs in no other IND, (iii) every *source* C_i lies outside L and is no copy, and (iv) if the IND has copies, its pointer neither contains $PK(R)$ nor lies inside it.

In particular, $\mathcal{K}$ excludes unique keys besides primary keys, copies of copies, and one-to-one replicas, whose pointer is the primary key. The next theorem fails if restriction (iv) is dropped in either way (Appendix A.3). For the schemas in $\mathcal{K}$, order-sufficiency reduces to a condition on each IND.

Theorem 4.4 (Exactness for $\mathcal{K}$). *Let Σ belong to $\mathcal{K}$. The following are equivalent: (1) Σ is order-sufficient under Fz; (2) Σ is order-insensitive under Fz; (3) $\mathcal{G} \models \Sigma$; (4) every IND $R[PD] \subseteq S[LC]$ in Σ satisfies $C \subseteq Fz$, or both $P \subseteq Fz$ and, for every position i, $C_i \in Fz$ or $D_i \in Fz$. Condition (4) is decidable in linear time.*

Unlike Thms. 3.5 and 4.2, Thm. 4.4 cannot be tested on updates with two changed entities. A violation of its condition may need four, as the following example shows.

Example 4.5 (A four-entity cycle). In Ex. 1.1, the foreign key is a projection of the cached-path IND. Merging the two INDs leaves the cached-path IND with pointer *gid*, copy *gpath* and source *path*, so the schema belongs to $\mathcal{K}$. Condition (4) holds exactly when *Group.path* is frozen, or when both *Project.gid* and *Project.gpath* are frozen. A workload that renames groups with projects modifies both *path* and *gpath* and violates both alternatives. Freezing only *gpath* violates them as well, although no copy then changes. Indeed, let p_1 move from g_1 to g_2 and p_2 from g_2 to g_1 while the two groups swap their paths, so that each project keeps its cached path. This update yields the precedence cycle $g_2 \to p_1 \to g_1 \to p_2 \to g_2$ of four entities.

4.2 Mutable Redundancy

The cycle of Ex. 4.5 needs two groups to exchange their paths, which fresh updates rule out. For fresh updates, class $\mathcal{K}$ separates the two notions.

Proposition 4.6 (Fresh updates on class $\mathcal{K}$). *Let Σ belong to $\mathcal{K}$. Every fresh modification update is executable under Fz if and only if every IND $R[PD] \subseteq S[LC]$ in Σ has $C_i \in Fz$ or $D_i \in Fz$ at every position i. Then writing the relations in a topological order of the IND graph, referencing relations first, executes every fresh modification update. Every fresh modification update is order-insensitive under Fz if and only if condition (4) of Thm. 4.4 holds. Otherwise, some fresh update with two changed entities is not executable, or not order-insensitive, respectively.*

Write order matters exactly when the first condition holds and the second fails. That is, a mutable pointer leads to a mutable source whose copy is frozen, and the referencing entities must move to another source before this source changes. In Ex. 4.5, freezing only *gpath* thus makes every fresh update executable, with projects written before groups. The updates of a single template are the modification updates under the freeze of every attribute that it does not modify. Hence Props. 3.7 and 4.6 decide templates one at a time and yield our central result.

Theorem 4.7 (Mutable redundancy). *Let Σ contain only FDs or belong to $\mathcal{K}$, and let a template only modify attributes and have fresh updates. The template needs atomic visibility if and only if it modifies (a) the right side A of an elementary FD $X \to A$ in Σ^+ whose left side X is not a superkey, or (b) both the source C_i and the copy D_i of an IND $R[PD] \subseteq S[LC]$ in Σ at some position i.*

In both cases the template rewrites a value and its copy, which is the test that our analyzer applies to code (Sec. 6.1). Unique keys besides the primary key, which $\mathcal{K}$ excludes, keep this test valid once the pointers that they pin are frozen (Prop. A.11 in Appendix A.3). The theorem treats FDs and class $\mathcal{K}$ separately. When no template modifies the right side of an FD whose left side is not a superkey, as in GitLab, where these right sides are frozen sharding keys (Sec. 6.4), case (a) is vacuous and case (b) decides the schema. The general combination is open.

4.3 Conditional INDs

Outside the exact classes, write order can help even modifications that exchange values. An IND into frozen reference data is one example (Ex. A.12 in Appendix A.3), and the one-way conditions on a status that applications maintain are another. A *conditional IND* (CIND) [15] $R[P; f \in A] \subseteq S[L; g \in B]$ requires, for every tuple t of R with $t[f] \in A$, a witness s of t under $R[P] \subseteq S[L]$ with $s[g] \in B$. Either condition may be absent. For example, let the groups and projects of Ex. 1.1 carry a flag *arch*. A project that is not archived must belong to a group that is not archived, that is, $Project[gid; arch \in \{0\}] \subseteq Group[gid; arch \in \{0\}]$. An update *enables* an entity of S if, for some CIND into S, the new tuple of the entity meets the condition on S but its old tuple does not. It *disables* the entity in the converse case. Here an inserted entity has no old tuple, and a deleted entity no new one. A nullable foreign key is the CIND whose condition on R is that the pointer is not null. It enables and disables nothing, and setting the pointer to null or to an existing parent changes one entity, so it needs no order at all. The class $\mathcal{C}$ consists of the schemas whose FDs are primary keys and whose INDs are CINDs with $L = PK(S)$ and an acyclic IND graph. Their CINDs into a relation S must all test the same attribute g of S, against sets B that form a chain under inclusion. On this class, enabled and disabled entities decide the write order.

Theorem 4.8 (Conditional INDs). *Let Σ belong to $\mathcal{C}$. (1) Every update, also one that inserts and deletes entities, is executable by writing first the entities that it enables, referenced relations first, then the other changed entities, and last those that it disables, referencing relations first. (2) Let every CIND with a condition on S have a pointer $P \notin PK(R)$, and share its pointer and S with no other CIND if it has no condition on R. Then every modification update is order-insensitive under Fz if and only if every CIND that tests a mutable attribute g has a frozen pointer P and tests no mutable attribute f.*

Part (1) extends the classical rule of Prop. 5.4 from insertions and deletions to every update, with enabling in place of insertion and disabling in place of deletion. Indeed, an enabled entity must become visible before the entities that come to need it, and a disabled entity after those that stop needing it. The chain of sets B prevents an update from both enabling and disabling one entity. Hence every schema of $\mathcal{C}$ is order-sufficient under every freeze. The restrictions of part (2) keep one CIND from implying another, which would make the condition sufficient but not necessary (Appendix A.3). In the example, archiving a group together with its projects writes the projects first and the group last. The schema is order-insensitive only if the workload never archives groups, or neither moves nor archives projects.

5 DESIGNING FOR ORDER

The characterization of Secs. 3 and 4 leaves a designer three decisions, which this section treats in turn: freezing attributes, choosing shard keys, and ordering the writes of each update.

5.1 Choosing the Frozen Attributes

To freeze an attribute that the workload modifies, a designer takes its modifications out of the workload. Each of them is either rewritten as re-keying, that is, as a deletion and an insertion ordered by their precedence graph (Sec. 5.3), or executed with atomic visibility, for example within one shard (Sec. 5.2). Both options have a cost, which we weight by the frequency of the templates that modify the attribute. The problem MIN-FREEZE takes a schema Σ, the set W of attributes that the workload modifies, weights on W, and a bound k. It asks whether freezing attributes of W of total weight at most k makes Σ order-sufficient.

With FDs only, Thm. 3.5 yields a unique minimum freeze for every weighting in polynomial time. Besides the primary keys, it contains the attributes on the left side of some FD of an elementary cover, and those that such an FD determines when its left side is not a superkey.

INDs make the choice combinatorial. For a schema in $\mathcal{K}$, the *conflict graph* H has the attributes in W as nodes. For every IND $R[PD] \subseteq S[LC]$, it has an edge between every attribute of the pointer P in W and every source of C in W, and an edge between C_i and D_i whenever both are in W. The edges of H are the pairs of attributes that cannot both stay mutable (Ex. A.13 in Appendix A.4).

Theorem 5.1 (Freezing as vertex cover). *Let Σ belong to $\mathcal{K}$. Freezing a set $Z \subseteq W$ makes Σ order-sufficient if and only if Z is a vertex cover of H. Every graph is the conflict graph of a schema in $\mathcal{K}$, so MIN-FREEZE is NP-hard already on $\mathcal{K}$.*

The hardness of Thm. 5.1 stems from odd cycles of the undirected IND graph. Indeed, every edge of H joins attributes of two relations that an IND connects. Hence H is bipartite whenever the undirected IND graph is, in particular when the foreign keys form a forest. By Thm. 5.1, our algorithm Freeze (Alg. 2 in Appendix A.4) only has to cover H. On a bipartite H, it computes a minimum-weight vertex cover as a minimum cut. Otherwise, it computes one of at most twice the minimum weight by the linear-time algorithm of Bar-Yehuda and Even [11]. If only fresh updates must be executable, it drops the edges at pointers (Prop. 4.6). Then H falls apart into stars of a source and its copies, and the cut takes the lighter side of every star.

5.2 Choosing Shard Keys

Def. 2.2 writes one entity at a time, but an engine can write all entities of one shard atomically. Hence a shard key relaxes the requirement, since entities in one shard may change together. We fix a shard key $SK(R) \subseteq Fz(R)$ for every relation R, so that no entity changes its shard. An update is *executable under SK* if its changed entities can be written in steps that each write entities of one shard atomically, such that the state after every step is legal. Since a hash function may map shard-key values to shards arbitrarily, we require this for every such map, one per co-location group (defined below). We call this requirement *hash placement*. We call an FD *unstable* if it is not in $\mathcal{G}$. Shard keys then preserve executability exactly under the following closure condition.

Theorem 5.2 (Admissible shard keys). *Let Σ contain only FDs, and let $SK(R) \subseteq Fz(R)$ for every relation R. Under hash placement, every modification update is executable under SK if and only if every unstable elementary FD $X \to A$ in Σ^+ satisfies $(X \cap Fz)^+ \supseteq SK$.*

Thm. 3.5 is the special case $SK = PK$. Only unstable FDs constrain the shard key, and the condition uses closure and not containment (Ex. A.14 in Appendix A.4). For fresh updates, the condition weakens to $X^+ \supseteq SK$ and applies only to the FDs that Prop. 3.7 retains. In the design of Ex. 3.6, where projects never move, the theorem admits only the group as the shard key of projects. This shard key turns every rename into a single-shard step (Ex. A.15 in Appendix A.4).

For INDs, entities that must become visible together need one shard, and only co-location puts entities of different relations on one shard. Hence we partition the relations into *co-location groups*, as Citus does [24], whose relations have shard keys of one length and share the map from shard-key values to shards. An IND $R[PD] \subseteq S[LC]$ *carries* the shard key if R and S share a group and Σ implies $t[SK(R)] = s[SK(S)]$ for every tuple t of R and its witness s. The chase decides this, since the IND graph is acyclic.

Theorem 5.3 (Admissible shard keys for $\mathcal{K}$). *Let Σ belong to $\mathcal{K}$, let $SK(R) \subseteq Fz(R)$ for every relation R, and let the relations be partitioned into co-location groups. Under hash placement, every modification update is executable under SK if and only if every IND of Σ that violates condition (4) of Thm. 4.4 carries the shard key.*

The equivalence of (1) and (4) in Thm. 4.4 is the special case $SK = PK$ with a group for every relation. That is, a reference that caches a mutable source must store the shard key of the entity that it references whenever its pointer or its copy can change as well. All other references may cross shards. For fresh updates, the condition is sufficient but not always necessary.

5.3 Compiling Write Orders

An executable update still has to be written in a legal order, and the engines differ in how an application can impose one. In this subsection, an update may also insert and delete entities, which we treat as entities whose old or new version is absent. Updates that only insert or only delete always admit a legal order when the INDs are acyclic, which justifies the classical rule.

Proposition 5.4 (Insertions and deletions). *Let the IND graph of Σ be acyclic. An update that only inserts entities is executable by writing the relations in reverse topological order of the IND graph, and an update that only deletes entities is executable by writing them in topological order.*

With a cyclic IND graph, two inserted entities may reference each other, and no order exists. Insertions and deletions are rarely order-insensitive, since a child written before its parent is an orphan. For them, write order is the remedy.

Fig. 2 traces this subsection on the template of Ex. 5.6. Updates that mix insertions, deletions and modifications, such as the re-keying of Sec. 5.1, need the precedence graph (Fig. 2a). The shards constrain their legal orders as follows.

Lemma 5.5 (Commit models). *Let a binary update and an assignment of its changed entities to shards be given. (1) If a shard may commit several times, the update can be executed by atomic single-shard steps if and only if every strongly connected component of $\mathcal{P}$ lies within one shard. (2) If every shard commits exactly once, in*

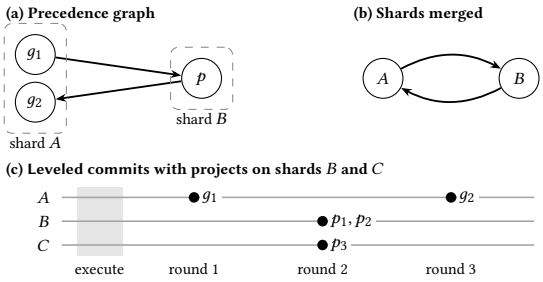


Figure 2: The Template of Ex. 5.6: Precedence, Merged Shards and Leveled Commits

Algorithm 1 Leveled Commits

Require: binary update, shards sh of changed entities, precedence graph $\mathcal{P}$, pause δ
Ensure: the update in the fewest rounds, or aborted before any write is visible
1: **for all** components C of $\mathcal{P}$ in topological order **do**
2: $\quad \lambda(C) \leftarrow \max(\{0\} \cup \{\lambda(C') + [sh(C') \neq sh(C)] : C' \to C\})$
3: $d \leftarrow 1 + \max_C \lambda(C)$
4: **for** $i \leftarrow 1, \ldots, d$ **do**
5: $\quad$ **for all** shards X **do**
6: $\quad\quad$ open $T_{i,X}$ on X for $\bigcup\{C : sh(C) = X,\ \lambda(C) = i - 1\}$
7: execute all writes in parallel, with no-wait locks and an outbox in every $T_{1,X}$
8: **if** a write fails **then**
9: $\quad$ abort every $T_{i,X}$ and **return**
10: **for** $i \leftarrow 1, \ldots, d$ **do**
11: $\quad$ **if** $i > 1$ **then**
12: $\quad\quad$ wait δ
13: $\quad$ commit every $T_{i,X}$ in parallel and await their commits

an order that the application chooses, the update can be executed if and only if merging the entities of every shard into one node, ignoring loops, yields an acyclic graph.

The engines of Sec. 2.1 realize these commit models differently. Single-shard transactions committed in a legal order, which we call ordered commits, realize model (1) on every engine. A single distributed transaction realizes model (2) only in part, since its shards commit in parallel on Citus and after the shard touched first on Vitess. It thus executes an update on Citus only if no edge of $\mathcal{P}$ joins two shards, and on Vitess whenever every such edge leaves the shard touched first. Model (2) is strictly more restrictive than model (1), as the following example shows.

Example 5.6 (Re-keyed rename). In Ex. 1.1, a template inserts a group g_1, moves a project p from group g_2 to g_1, and deletes g_2. It re-keys a rename of g_2 under a freeze of *path*. The foreign key gives the precedence chain from g_1 to p to g_2 of Fig. 2a. If g_1 and g_2 share a shard A while p lies on a shard B, merging the shards closes the cycle of Fig. 2b. Hence no single distributed transaction executes the template legally, whatever its statement order. Three single-shard transactions for g_1, p and g_2, committed one after the other, do.

Ordered commits that commit one entity at a time pay one durable commit per entity, but entities that no path of $\mathcal{P}$ orders may commit in parallel. An execution runs in *rounds* if the single-shard transactions of every round commit in parallel and in any order, and every round starts once the previous one has committed. The precedence graph and the shards fix the number of rounds.

Proposition 5.7 (Minimum rounds). *Let a binary update and an assignment of its changed entities to shards be given, such that every strongly connected component of $\mathcal{P}$ lies within one shard. For an entity e, let $\lambda(e)$ be the largest number of edges between two shards on a path of $\mathcal{P}$ that ends in e. Then the update can be executed in d rounds if and only if $d > \lambda(e)$ for every entity e. Committing in round i, for every shard, the entities e of that shard with $\lambda(e) = i - 1$ in one transaction attains this bound.*

We call these executions leveled commits. Alg. 1 computes λ in time linear in the size of $\mathcal{P}$ (lines 1–2) and groups the entities of every shard and value of λ into one transaction (lines 4–6). Their number of rounds depends on how often a path of $\mathcal{P}$ changes shards, and not on the numbers of entities or shards. For example, a parent with any number of children on other shards commits in two rounds, and the template of Ex. 5.6 in three, however many projects it moves (Fig. 2c). Every round consists of single-shard transactions, which the application sends in parallel straight to the nodes that hold their shards, whereas 2PC needs a node that coordinates the shards. Leveled commits protect the reader of assumption 5. If it sees the new tuple of an entity u, it also sees that of every v with an edge to u, since v commits in an earlier round or together with u and is read after u.

An application derives the input of leveled commits from its schema. For foreign keys, the edges of $\mathcal{P}$ follow the keys. An insertion writes a parent before its children, and a deletion removes the children before their parent. Re-keying writes the new parent, then the moved children, and removes the old parent last (Fig. 2a). We assume that the shard map, which places every entity by its shard key, does not change during an update. We also assume that no constraint checked within a shard relates writes of two rounds. Such a write would wait for an earlier round to commit, whereas Alg. 1 executes all writes first.

Leveled commits also confine the failures of an update. Alg. 1 opens the transactions of all rounds at once and executes their writes in parallel before the first round commits (lines 7–9). Hence a logical failure, such as a violated key, aborts the update before any write is visible. Only a crash can stop it between rounds, where it leaves a legal state. If no write of the update can fail logically, an *eager first round* commits already in the message that executes it, which saves one round trip. An update that must never stop between rounds can instead prepare all its transactions before the first round commits, an *ordered 2PC*, at the price of a log flush per transaction (Appendix B). To complete an update after a crash without a retry by the client, every transaction of the first round also stores the update in an outbox on its shard, together with the tuples that it read. From this outbox, a *relay* commits the remaining rounds. A relay that re-applies the new tuples blindly, as Sentry does, may overwrite a later update or move a child into a parent deleted in the meantime. Our relay validates first: before a round commits, it locks the entities of the round, reads the entities that share a constraint with them, and aborts unless these are as the previous round left them. Comparing values suffices, since legality depends on the values alone: an entity that another update changed and restored is as good as untouched. A crash may also leave a round partly committed, since its transactions commit independently. Such a

state is legal, as no edge of $\mathcal{P}$ joins two entities of one round, and writing the round again changes nothing for the entities that have committed. Every update that writes these entities, in one transaction or in rounds, must take the same locks. Then every state after a commit of the relay is legal (Prop. A.16 in Appendix A.4). The relay completes the update unless another update has overtaken it. In that case, the update stops in a legal state, and the outbox tells the application which writes remain. The client learns the outcome from the outbox as well, since the relay commits the later rounds without it.

Readers other than those of assumption 5, such as a join over several shards or a reader that follows a foreign key during a deletion, may read one shard after a later round and another before an earlier round. A pause between rounds separates such reads.

Proposition 5.8 (Grace period). *Let leveled commits send the commits of every round at least δ after all transactions of the previous round have acknowledged their commits. Then every reader that reads each shard once, all within less than δ, observes a legal state.*

That is, a reader faster than the pause observes a prefix of the rounds, and every such prefix is legal by Prop. 5.7. The pause needs no synchronized clocks, since the writer and the reader each measure a duration on their own clock. It adds δ to every round after the first but no message, and the client need not wait for it if the relay commits the later rounds. For example, if a parent is deleted one request timeout after its children, no request that reads a child and then its parent misses the parent.

Def. 2.2 considers one update at a time, whereas concurrent updates interleave their writes. Updates that change disjoint entities add no illegal state as long as their combination is binary and leaves a legal instance (Prop. A.19 in Appendix A.4). This is the condition of invariant confluence [4]. Other updates leave an orphan under ordered commits and under one distributed transaction alike. Such updates must lock the shared parent row, as the relay does. A later round waits for an earlier one in the application, where the deadlock detection of the engine cannot see the wait. Hence every lock request of leveled commits fails instead of waiting, and a failed update releases its locks and retries (Lem. A.17).

6 EXPERIMENTS

Our experiments answer four questions. (E1) How often does write order suffice in real applications, and how many of their constraints do the exact classes cover? (E2) Do the anomalies that write order prevents occur on Citus and Vitess, and does write order prevent them in less time than 2PC? (E3) How fast are the deciders, that is, our implementations of the tests of Thms. 3.5 to 5.2? (E4) How much does the design of Sec. 5 gain over the rules in use today? Questions (E1) and (E2) test the premise of (Q1) and (Q2): most real updates need write order rather than atomic visibility, and the engines do not supply it. Questions (E3) and (E4) test whether our answers are practical.

6.1 Setup

The application experiments study GitLab [35], whose 1,127 tables lie in three databases, and Sentry [64], whose 323 models lie in a control silo and in cells sharded by organization. The cross-database write templates of GitLab are the 15 transactions that it exempts

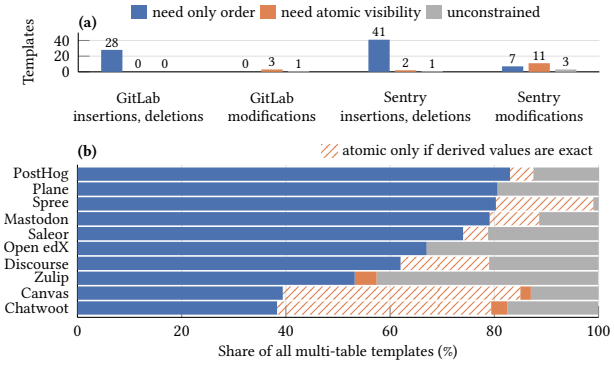


Figure 3: Templates by Requirement: GitLab and Sentry across Databases, by Hand (a), and Ten Further Applications across Tables, by the Analyzer (b)

from its guard against such writes and the deletions of 17 parents whose foreign keys cross databases. Those of Sentry are its 42 outbox replications and 23 synchronous remote writes. We classify every template by hand from the constraints between its rows. A template is unconstrained if no constraint relates its rows, needs only order if some write order keeps its constraints satisfied, and needs atomic visibility otherwise. A second annotator, who saw neither our labels nor the paper, labelled the templates from the same instructions and agrees with 88% of our labels ($\kappa = 0.71$), and on whether order suffices for 94 of the 97. The annotator doubts the other three only under concurrent updates, which the locks of Sec. 5.3 serialize, so we kept our labels, and the artifact holds both sheets. To test whether these findings carry over, we built a static analyzer, which reads the code of ten further open-source applications [16, 19, 26, 52, 55, 57, 58, 63, 67, 83] on Rails and Django without running it. It labels every template that writes several tables by Prop. 5.4 and Thm. 4.7, assuming that every table may lie on a shard of its own and that modifications write fresh values. The latter matters only for attributes that an FD or IND references, and the copied sources of GitLab and Sentry are paths, names and tokens, which are unique. We chose widely used applications on both frameworks: most keep the data of many tenants, such as accounts, realms or shops, and three already split their data across shards, databases or stores.

The engine experiments run Citus 14.2 on PostgreSQL 17 with a coordinator and three workers, and Vitess 24 in its 2PC mode on MySQL 8.4 with two shards. In the foreign-key runs, a writer inserts a parent and its child, and a reader that sees the child reads the parent in its next statement, which reverses the order of ordered commits. In the foreign-key runs on Citus, one worker commits with an injected delay, 10 ms unless stated otherwise, which separates the commit points of 2PC. In the copy runs, a writer updates a source and its copy in one transaction while a reader reads both in one statement.

The cost experiments insert a parent before its children, delete the children before their parent, or move the children to a new parent before deleting the old one, on 2 to 128 rows on alternating shards. Leveled commits send every transaction in one message

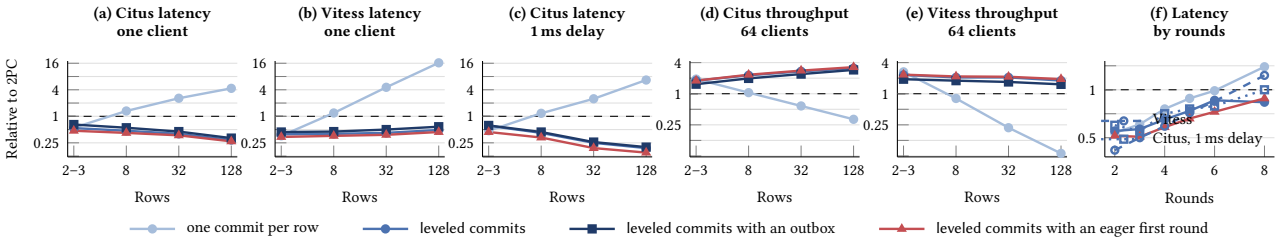


Figure 4: Cost of Leveled Commits, Which Are Not Atomic, Relative to the Fastest Two-Phase Commit

to the node of its shard. Their rounds follow each other without a pause, that is, $\delta = 0$ in Alg. 1, and only the pause runs of Sec. 6.3 add one. We compare them with ordered commits of one row each and with the fastest 2PC, which sends one statement per row or per shard and enters at the coordinator or at a worker. The clients write disjoint rows, so that no update waits for another, and proxies emulate network delay. Both engines commit durably, to one storage device that all nodes share. Fig. 4 reports median latency and throughput relative to the fastest 2PC, which takes 1.0 to 8.8 ms on Citus and 7.0 to 9.0 ms on Vitess from 2 to 128 rows. Each value is a geometric mean over the three templates and three repetitions, and its 95% interval lies within 12% of it in Figs. 4a to 4e.

All experiments ran on one server with 1 TB of memory and two 64-core AMD EPYC 7742 processors, and the artifact contains the scripts and the full protocol.

6.2 How Often Does Order Suffice?

We first ask whether write order is the remedy that real applications need (E1). Fig. 3a shows that most cross-database write templates need only order in both applications, 28 of 32 in GitLab and 48 of 65 in Sentry. Order fails almost only for modifications. Every template that needs atomic visibility maintains redundant data whose copy or derivation lives on another database, such as the full paths of routes in GitLab and the tokens that Sentry copies into its replicas. For each of these 16 templates, 3 in GitLab and 13 in Sentry, an audit of the code finds a reader that relies on the copy agreeing with its source. For 13 of them, the reader decides authentication or authorization, such as a revoked token that still authenticates. Where order helps a modification, Thm. 4.8 explains why. Five of the seven modifications of Sentry that need only order keep one-way conditions, such as an integration that is disabled after its repositories. None of them is order-insensitive, and the other two also insert or delete rows. Thm. 4.7 thus shows in the exceptions rather than in the rule. The order-only templates insert, delete or move pointers, which the classical rule explains, or keep one-way conditions. A modification that rewrites no copy relates no rows across databases, so it is unconstrained rather than order-only. Hence order is the common case, counting each template once since frequencies are not public, and the exceptions show in the schema, as Thm. 4.7 predicts.

The exact classes of Sec. 4 cover 22 of the 32 templates of GitLab and 41 of the 65 of Sentry (Appendix B).

Fig. 3b shows that the analyzer finds the same pattern in the ten further applications: 99% of their 5,049 constrained templates need

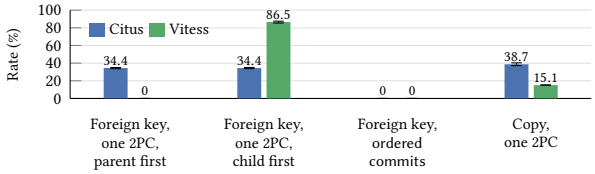


Figure 5: Orphans per Run and Inconsistent Copies per Read, with 95% Confidence Intervals and a 10 ms Commit Delay on One Citus Worker in the Foreign-Key Runs

only order if derived values may lag behind their rows. Of these 4,998 templates, 681 only insert or delete rows, 286 move pointers along references, and the other 4,031 modify further attributes next to insertions or deletions. This share is an upper bound. On the 97 templates of GitLab and Sentry, the analyzer finds 72 and agrees with 57 of our labels on them. It calls 9 order-only templates unconstrained and misses the copies of 4 templates that need atomic visibility, but 11 of its 12 verdicts of atomic visibility are right. Blind hand labels of 135 templates sampled from the ten applications put the share that needs only order at about 82% (95% interval 74–90%). The analyzer agrees with 79 of these labels. It calls 40 templates order-only that the labels find unconstrained and misses 16 that need atomic visibility, mostly copies that it cannot parse, whereas the labels confirm all five of its sampled verdicts of atomic visibility. Derived values such as counters lie outside our model, since they are neither FDs nor INDs. If they must stay exact, the share drops to 78%.

However, write order protects only the readers of assumption 5. An audit of the read paths of the 76 order-only templates shows that most readers fit it or tolerate the rest. Write order alone protects every reader of 47 of these templates, whose readers reverse the write order or read one side. Each of the other 29 has a reader that reads against the write order, mostly one that loads the parent of a child during a deletion. Both applications already expect such parents to be missing, since their own order leaves dangling children to a background task. Sentry tolerates a missing parent at 80 of the 88 places where its code follows such a reference. For 9 templates such a reader raises an error, which we reported to both projects.

6.3 Does Write Order Matter on Real Engines?

Having seen that most templates need only order, we ask whether 2PC provides it, and what leveled commits cost (E2). Fig. 5 shows that one 2PC exposes the anomalies that write order prevents,

whereas ordered commits prevent them on both engines. On Vitess, which commits the shard touched first before the others (Sec. 2.1), a reader saw an orphan in 86% of the runs that wrote the child first and in none of those that wrote the parent first. On Citus, the commit points of the workers lie so close together that only 0.1% of the runs exposed an orphan without delay. Once one worker commits 1 ms later or more, a reader saw an orphan in almost every run whose parent lay on that worker, whatever the statement order. That is, one 2PC exposed an orphan in 34% of the runs from 1 ms up to the 10 ms of Fig. 5. Two ordered commits of the parent and the child produced no orphan on either engine. Hence write orders need separate transactions on Citus but only the statement order on Vitess (Lem. 5.5). For the templates that need atomic visibility, 2PC is no remedy either. A transaction that wrote a source and its copy exposed an inconsistent pair in 15% of reads on Vitess and 39% on Citus.

Figs. 4a to 4c show that leveled commits, which provide these separate transactions, take less time than the fastest 2PC, however many rows a template writes, also behind added delay. At 128 rows they take less than a third of its time on Citus and half of it on Vitess, and a fifth behind proxies that add 1 ms to every round trip on Citus. Most of this advantage stems from the access path that single-shard transactions allow (Sec. 5.3). On Citus with one client, the phase that executes the statements before any commit holds 78% of the gap to the fastest 2PC at 8 rows and 95% at 128 rows. Indeed, leveled commits send the statements of every shard in one message to the node that holds it, whereas 2PC routes every statement through the coordinator, even when it enters at a worker.

The cost of write order lies in its rounds and not in its rows. Ordered commits of one row each take four times as long as 2PC on Citus and sixteen times on Vitess at 128 rows, since they pay one durable commit per row. Leveled commits instead need the fewest rounds (Prop. 5.7). The number of rounds, in turn, bounds the advantage of leveled commits. To vary the rounds, we insert paths of 2 to 8 rows whose every edge joins two nodes, so that leveled commits need one round per row. Fig. 4f shows that the advantage shrinks as rounds are added: it holds up to eight rounds on Citus, but only up to six on Vitess and behind 1 ms proxies, the bound that holds in every configuration. Indeed, every round adds a round trip and a durable commit, whereas 2PC pays its prepare and commit rounds once.

The advantage carries over from latency to throughput. With 64 clients, leveled commits sustain 1.8 to 3.1 times the throughput of 2PC at every size on both engines (Figs. 4d and 4e).

Recovery and locks cost little (Appendix B). An outbox raises the latency of leveled commits by at most a fifth. When a proxy dropped commit messages to a worker, one 2PC left orphans until the periodic recovery of Citus ran, whereas leveled commits with an outbox left none. The relay of Prop. A.16 left none either in 3,000 recoveries that raced with a conflicting update, where a relay that re-applied the new tuples left 653. Locks on shared parents prevent the orphans of conflicting updates and cost an addition 0.14 ms. Hence leveled commits sent straight to the shards protect the readers of assumption 5 in less time than one 2PC, also when commit messages are lost or updates conflict.

Read direction decides whether write order alone protects a reader. Fig. 6 shows that, during deletions under ordered commits,

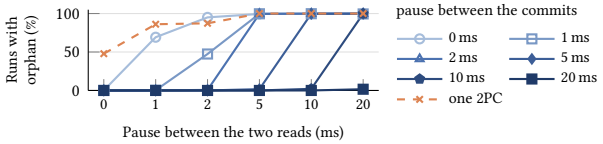


Figure 6: Orphans during Deletions by Pause between the Reads and between the Commits

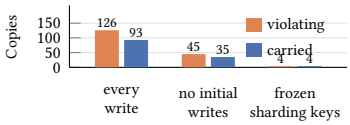


Figure 7: Copies of GitLab Violating Condition (4) of Thm. 4.4, and Those Carried by Co-Location

a reader that follows the foreign key from the child saw an orphan in 69% of the runs once 1 ms separated its two reads. A reader that lists the children of the parent saw none in 6,000 runs. Indeed, the former sees an orphan whenever its two reads enclose both commits, which no order prevents, and insertions behave alike with the two readers swapped. One 2PC exposes this reader more often, in 48% of the runs even without a pause, since its two commits fall more easily between the two reads.

A pause between the two commits closes this window for every reader that reads faster than the pause, as Prop. 5.8 predicts. In Fig. 6, the orphans of each pause appear only once the reads lie further apart than the commits. Among 700,952 pairs of reads that took no longer than the pause, in deletions and insertions alike, none saw an orphan. Hence write order with a pause protects every reader that finishes within the pause, and only slower readers must tolerate missing parents, as those of Sentry do (Sec. 6.2).

6.4 How Fast Is Design, and What Does It Gain?

Having seen that write order matters on real engines, we ask whether a design tool can decide order-sufficiency fast (E3), and what the design of Sec. 5 gains over the rules in use today (E4). Decision time grows linearly with the schema. Every decider answers within 4 s on random schemas with 10,000 relations and within 0.4 s on the 1,127 relations of GitLab, and the minimum cut finds the optimal freeze over 70,000 attributes in 0.6 s (Appendix B).

Thm. 5.3 explains why GitLab can shard by its declared keys without atomic visibility. GitLab maintains 903 copies that its schema or its code exhibits, 808 of sharding keys and 95 of other attributes that callbacks write. Fig. 7 shows that, once the sharding keys are frozen, none of the 808 violates condition (4) of Thm. 4.4 and co-location on the declared keys carries the 4 attribute copies that do. GitLab modifies sharding keys only when it transfers, imports or backfills data, such as a project that moves to another organization. Counted as modifications, these templates raise the violations to 45, of which co-location carries only 35, so they must be re-keyed or run with atomic visibility. For fresh updates, no FD of GitLab needs atomic visibility at all, since its 671 FDs whose left sides are no keys copy frozen sharding keys. Optimal freezing pays off

once templates modify attributes at different frequencies: under synthetic log-normal frequencies, it saves a median 36 to 70% of the weight over freezing every copied source (Appendix B).

7 RELATED WORK

Coordination avoidance. Invariant confluence [4] tells whether an operation can run without coordination. Indigo [10], IPA [9] and the proof rule of Gotsman et al. [36] decide which operations of a replicated database must synchronize to preserve invariants, or repair them so that they need not. RedBlue consistency [46] and the homeostasis protocol [62] let the operations that preserve the invariants run without coordination, and Salt [76] and Callas [77] confine coordination to the transactions that need it. Robustness analyses decide whether transaction templates run serializably under snapshot isolation [12, 17, 30] or read committed [40, 71]. Korth and Speegle [41], like Rastogi et al. [59], defined correctness by the consistency constraints instead of serializability. These analyses treat the writes of one operation or transaction as atomic, whereas we split the constraints that need coordination into those that need only order and those that need atomic visibility.

Write ordering. Soft updates [31] order metadata writes so that a file system stays consistent after a crash, and object-relational mappers sort the rows of a unit of work topologically, both along a fixed dependency rule. The cascades of Rails (`dependent: :destroy`) and Django (`on_delete`) delete children before their parents by the same rule, but within one transaction, whose statement order protects no reader on Citus (Lem. 5.5). Semantic consistency [32] interleaves transactions that the application declares compatible, and multilevel atomicity [50] extends it to a hierarchy of breakpoints. In multidatabases, commitment ordering [60] and quasi serializability [27] make every local database order the global transactions alike. Sagas [33] commit the steps of a long transaction separately and compensate on failure, and the transactional outbox [61] makes such steps reliable, but both leave the consistency of the intermediate states to the application. Transaction chains [80] also commit the hops of a transaction one shard at a time. However, they derive the admissible chains from a static analysis of all transactions and promise serializability to them, whereas leveled commits derive the order from the schema and promise legal states to readers that no analysis sees. Read-copy update [53] frees a removed element only after every reader that may still reference it has finished, as a pause between rounds does for readers of bounded duration. Markowitz [51] gave conditions under which referential actions have the same effect in every order of enforcement. Deferrable constraints of SQL postpone checks only to the commit of one transaction. We are unaware of previous work that characterizes when a legal write order exists for the constraints of a schema, or how schema and placement decide it.

Schema design. Normalization targets redundancy, with normal forms such as the elementary key normal form [79], the object normal forms of Biskup [14], and normal forms parameterized by the numbers of minimal keys and FDs [81, 82]. Redundancy has been characterized semantically [73] and information-theoretically [2]. LeDoux and Parker [43] related BCNF to the anomalies of given updates. Levene and Vincent [44] justified the inclusion dependency normal form, whose INDs are acyclic and reference keys as in our class $\mathcal{A}$, by the absence of redundancy and update anomalies. These works consider a database whose updates are atomic, whereas we ask, relative to the frozen attributes of a workload, which constraints survive the intermediate states of updates on sharded engines. For fresh updates, our answer is again redundancy, but only redundancy among mutable attributes. Update-stable consequences are related to the independence of queries from updates [28, 45], as no modification of the workload can violate them. Vianu [72] constrained how tuples evolve with dynamic FDs, whereas frozen attributes only fix the attributes that never change. Bravo et al. [15] introduced conditional INDs for data cleaning and studied their implication, whereas we ask which write orders keep them satisfied.

Sharded engines and practice. Helland [38] advocated designs without distributed transactions. In practice, applications enforce many integrity constraints themselves [5], maintain constraints across microservices by hand [42] and coordinate operations with ad hoc transactions [69]. Schism [25] and Horticulture [56] partition a database so that few transactions of a workload span partitions, whereas Thm. 5.2 asks which shard keys keep the remaining updates free of atomic visibility. F1 [66] interleaves child rows under their root and PREF [78] co-locates rows along foreign keys, which Thm. 5.3 requires only for the INDs that need atomic visibility.

8 CONCLUSION

We have studied which updates on sharded SQL engines need atomic visibility and when write order alone keeps the constraints of a schema satisfied. For (Q1), under FDs and foreign keys with cached columns, a fresh update needs atomic visibility exactly when it rewrites a value and its copy. For arbitrary modifications, the update-stable consequences decide order-sufficiency exactly for FDs, where they recover the object normal form of Biskup, and for three classes of INDs. Conditional INDs of class C never need atomic visibility, but they need write order. For (Q2), choosing the frozen attributes is NP-hard but a minimum cut when foreign keys with cached columns form a forest, and the frozen attributes decide the admissible shard keys. The remaining write orders compile to leveled commits, which need the fewest rounds and, sent straight to the shards, take less time than 2PC for up to six rounds. In GitLab, Sentry and ten further applications, most write templates across databases or tables need only order, and almost all others maintain copies or derived values of mutable attributes.

Future work will address derived values, conditional INDs outside C, and acyclic INDs outside our three classes. Overall, atomic visibility is the price of mutable redundancy, and write order is a design decision that the schema can justify.

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A PROOFS

This appendix proves the results of Secs. 3 to 5 in the order in which they appear. Throughout, instances are finite, and implication and the closure Σ^+ are taken in the finite sense. For an instance I and an entity e, we write $I(e)$ for the tuple of e in I.

A.1 Binary Updates

An update that inserts or deletes entities shows an inserted entity in I, and a deleted one in J, with an absent version, which violates no FD, needs no witness and witnesses nothing.

Proposition A.1 (Binary updates). *Let every IND $R[X] \subseteq S[Y]$ in Σ have a key K of S inside Y, and let no entity agree on K in J with another entity in I. Then the update is binary, and its precedence graph has an edge from v to u exactly when (i) the new tuple of u and the old tuple of v violate an FD in Σ, (ii) the new tuple of u has a witness under some IND only in the new tuple of v, or (iii) the old tuple of v has a witness under some IND only in the old tuple of u.*

Proof. An intermediate state shows every changed entity in its old or its new version, and every other entity in its only version. An FD relates two tuples, and two versions that violate it are not both old, since I is legal, nor both new, since J is legal, nor do they involve an unchanged entity, since its version occurs in I and in J. Hence the state satisfies the FDs exactly when it shows no pair of case (i). Consider an IND $R[X] \subseteq S[Y]$, a version t of an entity e, and two versions s and s' of entities w and w' with $s[Y] = s'[Y] = t[X]$. Then s and s' agree on K, so $w = w'$ by the hypothesis if one of them is old and the other new, and by the key K of I or J otherwise. Thus all candidate witnesses of t are versions of one entity w. If t is new, some version of w is a witness, since J is legal, and the state fails the IND at t only if it shows the old version of w while only the new one is a witness, which is case (ii) with $u = e$ and $v = w$, as w then changes. Symmetrically, if t is old, the state fails the IND at t only in case (iii) with $v = e$ and $u = w$. Hence a state is legal exactly when it shows no forbidden pair, that is, when no edge leads into the set of entities that it shows new from outside. □

The hypothesis is needed. If an update deletes an entity w' of S and inserts an entity w with the same value of K, while a changed entity e of R starts to reference this value, then the state that shows e new and w' deleted is illegal, although the states with only one of them are legal, and no precedence graph expresses this. For modification updates, the hypothesis holds whenever $K \subseteq Fz(S)$ or the update is fresh (Sec. 3.3), and inserted entities meet it when they take values of K that I does not contain.

On the running example, Prop. A.1 reads as follows.

Example A.2 (Moves and renames). Consider again Ex. 1.1. Moving a project changes a single entity, so every move is executable. Renaming a group g with a project p is not executable, since the old g is no witness of the new p for the cached-path IND, nor the new g of the old p. By Prop. A.1, the update is binary, with precedence edges from g to p and back. Hence g and p must become visible together, and the schema is not order-sufficient for this workload.

A.2 Proofs for Sec. 3

The running example illustrates the update-stable consequences of Sec. 3.1.

Example A.3 (Unstable cached paths). In Ex. 1.1, $M^* = Fz$ because *Project.gid* is not frozen. The foreign key belongs to $\mathcal{G}$ because its right side *gid* is frozen. The cached-path IND does not, because its right side contains the mutable attribute *path* and its left side is mutable. Hence $\mathcal{G}$ misses exactly the constraint that a rename violates. If *Project.gid* and *Group.path* were frozen, M^* would also contain *Project.gpath*, since it copies the frozen path of the group that the frozen pointer *gid* identifies.

We start with the invariant of the frozen closure.

Proof of Lem. 3.2. Let (I, J) be a modification update. We show by induction on the construction of M^* as a least fixpoint that $I(e)$ and $J(e)$ agree on $M^*(R)$ for every entity e of every relation R. The base case Fz holds by Def. 2.1. For the inductive step, let the rule add X_i to $M^*(R)$ through an IND $R[X] \subseteq S[Y]$ in Σ^+, where Y contains a key K of S with $K \subseteq M^*(S)$ and the attributes X_K of X at the positions of K lie in $M^*(R)$. Let e be an entity of R, and let $u = I(e)$ and $u' = J(e)$. Since I and J satisfy the IND, there are entities w and w' of S with $I(w)[Y] = u[X]$ and $J(w')[Y] = u'[X]$. By the induction hypothesis, $u[X_K] = u'[X_K]$ and $I(w)[K] = J(w)[K]$. Then $J(w)[K] = u[X_K] = u'[X_K] = J(w')[K]$, and $w = w'$ because J satisfies the key K. Consequently, $u'[X_i] = J(w)[Y_i] = I(w)[Y_i] = u[X_i]$ whenever $Y_i \in M^*(S)$. □

The invariant yields the claim made after Def. 3.3.

Lemma A.4. *Every constraint in $\mathcal{G}$ holds in every intermediate state of every modification update.*

Proof. Let (I, J) be a modification update, and let $L = I \oplus T$ for some $T \subseteq Ch$. Every tuple of L is $I(e)$ or $J(e)$ for an entity e, and $I(e)$ and $J(e)$ agree on M^* by Lem. 3.2. Consider first an FD $X \to A$ in $\mathcal{G}$, and let the tuples of two distinct entities e and f agree on X in L. If X contains a key $K \subseteq M^*$, then $I(e)$ and $I(f)$ agree on K, which contradicts the key K in I. If $X \cup \{A\} \subseteq M^*$, then $I(e)$ and $I(f)$ carry the values of the two tuples on $X \cup \{A\}$, so the two tuples agree on A because I satisfies the FD. Consider next an IND $R[X] \subseteq S[Y]$ in $\mathcal{G}$, and let t be the tuple of an entity e of R in L. If $Y \subseteq M^*(S)$, then I, J and L have the same projection on $S[Y]$, and $t[X]$ lies in the projection of I or of J. Otherwise, $X \subseteq M^*(R)$, and Y contains a key K of S with $K \subseteq M^*(S)$. Then $t[X] = I(e)[X] = J(e)[X]$, and there are entities w and w' of S with $I(w)[Y] = t[X] = J(w')[Y]$. By Lem. 3.2, $J(w)[K] = I(w)[K] = J(w')[K]$, so $w = w'$. The tuple of w in L is a witness for t. □

Proof of Thm. 3.4. Every intermediate state is a finite instance that satisfies $\mathcal{G}$ by Lem. A.4. Since $\mathcal{G}$ implies Σ, the state satisfies Σ. Hence every modification update is order-insensitive, and in particular executable. □

For FDs we use a classical fact: an instance of a relation with two tuples satisfies a set of FDs if and only if the attributes on which the tuples agree, their *agree set*, form a closed set. For a relation R with attributes U, we write $N = U \setminus Fz(R)$ for its mutable attributes. Two tuples t_1 and t_2 *swap their mutable parts* if they are replaced by

t_1', which agrees with t_1 on Fz and with t_2 on N, and by t_2', defined symmetrically. The agree set of t_1' and t_2' is that of t_1 and t_2, and the agree set of t_1' and t_2 is the union of N and the frozen part of the agree set of t_1 and t_2. The following lemma justifies the remark after Thm. 3.5 that order-sufficiency freezes every key.

Lemma A.5. *Let Σ contain only FDs and satisfy condition (4) of Thm. 3.5. Then every minimal key of every relation is frozen.*

Proof. Assume that a minimal key K of a relation R has a mutable attribute, and let $C = (K \cap Fz)^+$. Since K is minimal, $C \neq U$, and therefore $PK \not\subseteq C$. Let $Z = (C \cap Fz) \cup N$. Then $Z \supseteq K$, so $Z^+ = U$, whereas $PK \not\subseteq Z$ because $PK \subseteq Fz$. Consequently, Z is not closed, and some elementary FD $Y \to b$ in Σ^+ has $Y \subseteq Z$ and $b \notin Z$. By condition (4), $Y \subseteq Fz$, so $Y \subseteq C$ and $b \in C$. Since $N \subseteq Z$, also $b \in Fz$, and thus $b \in C \cap Fz \subseteq Z$, a contradiction. □

Proof of Thm. 3.5. Since Σ contains no INDs, $M^* = Fz$.

(4)⇒(3). Let $X \to A$ be an elementary FD in Σ^+. If X is a superkey, it contains a key that determines A, and since no proper subset of X determines A, X is itself a key. By (4), X is a frozen key, so the FD belongs to $\mathcal{G}$. Otherwise, $X \cup \{A\} \subseteq Fz$ by (4), and the FD belongs to $\mathcal{G}$ again. Every nontrivial FD of Σ follows from an elementary FD of Σ^+ with a smaller left side, so $\mathcal{G} \models \Sigma$.

(3)⇒(2) is Thm. 3.4, and (2)⇒(1) holds because every prefix state is an intermediate state.

(1)⇒(4). Let $X \to A$ be an elementary FD over R in Σ^+ that violates (4). We build a modification update (I, J) in which R holds two tuples, every other relation is empty, and either intermediate state with one changed entity violates Σ. In each case, I holds two tuples whose agree set C is closed and differs from U, so they differ on PK and are distinct entities. *Case 1: X is a superkey.* Then X is a minimal key with a mutable attribute, as shown above. Let $C = (X \cap Fz)^+$, and let J arise by swapping the mutable parts. Both entities change, since a mutable attribute of X lies outside C by the minimality of X. The intermediate state $\{t_1', t_2\}$ agrees on $(C \cap Fz) \cup N \supseteq X$ but not on PK, which violates the key X. *Case 2: X is not a superkey and A is mutable.* Let $C = X^+$, and let J give both tuples the same new value on A. The agree set of J is C again, and that of $\{t_1', t_2\}$ is $C \setminus \{A\} \supseteq X$, which violates $X \to A$. *Case 3: X is not a superkey, A is frozen, and some $x \in X$ is mutable.* Let $C = (X \setminus \{x\})^+$. Since the FD is elementary, $A \notin C$ and $x \notin C$. Let J arise by swapping the mutable parts, which changes both entities because $x \notin C$. The intermediate state $\{t_1', t_2\}$ agrees on $(C \cap Fz) \cup N \supseteq X$ but not on the frozen attribute A. The three cases cover every violation of (4), and the intermediate state $\{t_1, t_2'\}$ is symmetric in each. Hence the update is not executable. □

The proof also yields the polynomial test claimed in Thm. 3.5. For a set Z of attributes of R, let $Z' = Z \cup ((Z \cap Fz)^+ \cap Fz)$. The closure of Z under $\mathcal{G}$ is U if $(Z \cap Fz)^+ = U$, and Z' otherwise. Indeed, in the first case $Z \cap Fz$ contains a frozen key, and in the second case Z' is closed under both kinds of FDs in $\mathcal{G}$, since $(Z' \cap Fz)^+ = (Z \cap Fz)^+$ contains no key. Deciding $\mathcal{G} \models \Sigma$ takes one closure under Σ for every FD of Σ.

For fresh updates, we use that the left sides of elementary FDs consist of attributes that occur on left sides of Σ.

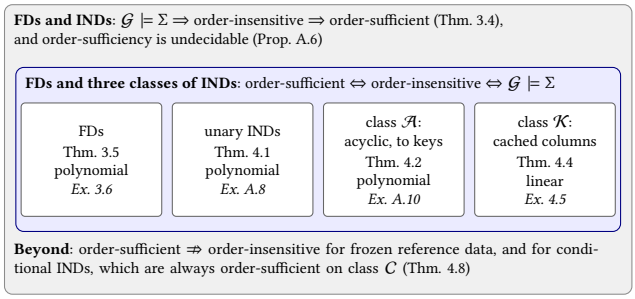


Figure 8: Order-Sufficiency by Class of Constraints

Proof of Prop. 3.7. Let $X \to A$ be an elementary FD in Σ^+, and let $x \in X$ occur on no left side of an FD in Σ. Then x fires no FD of Σ, so the closure of X is the closure of $X \setminus \{x\}$ together with x, and $A \neq x$ lies in the closure of $X \setminus \{x\}$, which contradicts elementarity. Hence every attribute of X receives only new values from a fresh update, unless the update leaves it unchanged.

Assume the condition, and let the tuples of two distinct entities violate Σ in an intermediate state $I \oplus T$. They violate an elementary FD $X \to A$ of Σ^+, and since I and J satisfy Σ, one of them is the new tuple $J(u)$ of some $u \in T$ and the other the old tuple $I(v)$ of some $v \in Ch \setminus T$. For every $x \in X$, the update leaves $J(u)[x] = I(u)[x]$, since a new value of x would differ from $I(v)[x]$. Hence $I(u)$ and $I(v)$ agree on X, so X is no superkey, and they agree on A, so $J(u)[A] \neq I(u)[A]$ and A is mutable, which contradicts the condition. Consequently, every fresh update is order-insensitive.

Conversely, let $X \to A$ be an elementary FD in Σ^+ whose left side is no superkey and whose right side is mutable. Case 2 of the proof of Thm. 3.5 takes two tuples whose agree set is X^+ and gives both a value of A that occurs nowhere in I. The update writes only this value, so it is fresh, and neither intermediate state with one changed entity satisfies $X \to A$. □

A.3 Proofs for Sec. 4

Fig. 8 summarizes the results of Sec. 4 by class of constraints.

Order-sufficiency inherits the undecidability of implication.

Proposition A.6 (Undecidability). *For FDs and INDs, it is undecidable whether Σ is order-sufficient under Fz, and whether Σ is order-insensitive under Fz, already when the workload modifies a single attribute and the frozen attributes are exactly the prime attributes.*

The proof needs a source problem without FDs whose left side is empty.

Proof of Prop. A.6. Finite implication of an FD by FDs and INDs remains undecidable when every FD has a nonempty left side, as Abiteboul et al. note in their proof of this result [1, p. 200]. Let Σ be such a set over relations with *original* attributes, and let $X \to A$ be an FD over R with $A \notin X$. We may assume $X \neq \emptyset$. Indeed, the instance that holds one tuple of zeros in every relation satisfies Σ, and so does its disjoint union with a copy of ones, since FDs with nonempty left sides and INDs survive disjoint unions. Hence Σ implies no nontrivial FD with an empty left side, and the restriction to $X \neq \emptyset$ keeps the problem undecidable.

We add to every relation S a surrogate primary key ID_S, add to R a new attribute V with the FD $XV \to A$, and freeze every attribute except V. Call the result Σ'. Assume first that $\Sigma \models X \to A$. Every intermediate state L of a modification update (I, J) agrees with I outside V. It satisfies every constraint of Σ' that does not mention V, as I does, and its projection on the original attributes satisfies Σ and therefore $X \to A$. Consequently, L satisfies $XV \to A$, and Σ' is order-insensitive. Assume next that $\Sigma \not\models X \to A$, and let I_0 be a finite instance of Σ with tuples t_1 and t_2 that agree on X but not on A. We extend I_0 with distinct surrogate values, give t_1 and t_2 the V-values 1 and 2, and give every other tuple a new V-value. The result I satisfies Σ', and so does the instance J that swaps the V-values of t_1 and t_2. Either intermediate state with one changed entity makes t_1 and t_2 agree on XV but not on A. Hence Σ' is order-sufficient, and order-insensitive, exactly when $\Sigma \models X \to A$, which shows the first claim.

For the second claim, we add to every relation S and every original attribute B of S a frozen attribute N_B with the FD $BN_B \to ID_S$, and to R the FDs $XN_C \to ID_R$ for every original attribute C of R. The argument above carries over, since the new FDs do not mention V and pairwise distinct N-values satisfy them. It remains to show that the prime attributes are exactly the frozen ones. The surrogate ID_S is a key. Every original B and its N_B form a minimal key, for two reasons. Firstly, B alone is no superkey, as witnessed by two tuples that agree on all original attributes and on V and differ elsewhere. Secondly, N_B alone is no superkey, as witnessed by the disjoint union above, extended with equal N-values and distinct V-values: its two tuples per relation differ on every original attribute, which satisfies the new FDs because $X \neq \emptyset$. Finally, V is not prime. The first witness shows that every superkey contains a surrogate or an N-attribute. Let K be a minimal key of R with $V \in K$. Then K contains no surrogate, since the surrogate alone is a key, and hence some N_C. Since $K \setminus \{V\}$ is no superkey, some instance has tuples $p \neq q$ that agree on $K \setminus \{V\}$, and hence on N_C. By $XN_C \to ID_R$, the tuples p and q differ on X. Giving every tuple of R a distinct V-value, except that q takes the value of p, preserves $XV \to A$ and every other constraint. The resulting instance has two tuples that agree on K, a contradiction. □

For unary INDs, the key step permutes values column by column.

Lemma A.7. *Let Σ consist of FDs and unary INDs, let I satisfy Σ, and let R be a relation. For every mutable attribute B of R, let π_B be a permutation of the values in column B of $I(R)$. Replacing every tuple t of R by the tuple that agrees with t on $Fz(R)$ and takes the value $\pi_B(t[B])$ on every mutable B yields an instance J, and (I, J) is a modification update.*

Proof. Entities and frozen values are unchanged. Every π_B is injective, so every two tuples of R keep their agree set, and J satisfies the FDs. Every column keeps its set of values, and J satisfies the unary INDs. □

Proof of Thm. 4.1. Since $Fz \subseteq M^*$, a unary IND with a frozen right side belongs to $\mathcal{G}$.

(4)⇒(3). The first part of (4) places every nontrivial unary IND of Σ^+ in $\mathcal{G}$. The second part places every elementary FD of Σ^+ in $\mathcal{G}$, as in the proof of Thm. 3.5. Hence $\mathcal{G} \models \Sigma$. Thm. 3.4 gives (3)⇒(2), and (2)⇒(1) is immediate. (1)⇒(5) holds because every write order passes through a prefix state that contains the new tuple of exactly one of two such entities.

(5)⇒(4). Assume that (4) fails. In each of the following cases, we take a finite instance I of Σ, obtain J by Lem. A.7, and exhibit two changed entities that must become visible together. Every intermediate state that contains the new tuple of exactly one of them violates a constraint of Σ^+, and hence Σ. We describe the state in which the first entity is new, and the other state is symmetric. We use that the disjoint union of an instance of Σ with a copy in which all values are renamed is again an instance of Σ, since the FDs have nonempty left sides. *Case 1: a nontrivial unary IND $R[A] \subseteq S[B]$ in Σ^+ has a mutable right side.* Let I be the disjoint union of the instance with one tuple of zeros per relation and its copy with $0'$. Transpose 0 and $0'$ in column $S.B$. The changed entities are the two tuples of S. If only the first is new, column $S.B$ holds only $0'$, whereas column $R.A$ still holds 0. *Case 2: some elementary FD $X \to A$ in Σ^+ has a superkey X with a mutable attribute.* Then X is a minimal key, and $X \cap Fz$ is no superkey. Let I hold distinct tuples t_1 and t_2 of R that agree on $X \cap Fz$, and transpose $t_1[B]$ and $t_2[B]$ in every mutable column B. Both tuples change, since they differ on a mutable attribute of the key X. The new t_1 agrees with the old t_2 on X but not on PK, which violates the key X. *Case 3: some elementary FD $X \to A$ in Σ^+ has a non-superkey X and a mutable A.* Let I hold distinct tuples t_1 and t_2 that agree on X, and hence on A with a value a, together with a renamed copy that carries a value b in column A. Transpose a and b in column A. The new t_1 agrees with the old t_2 on X but not on A. *Case 4: some elementary FD $X \to A$ in Σ^+ has a non-superkey X with a mutable x, and A is frozen.* Since $X \setminus \{x\}$ does not determine A, let I hold tuples t_1 and t_2 that agree on $X \setminus \{x\}$ but not on A, and transpose $t_1[B]$ and $t_2[B]$ in every mutable column B. Both tuples change, since they differ on x, as X determines A. The new t_1 agrees with the old t_2 on X but not on A. The four cases cover every violation of (4).

Polynomial time. By the axiomatization of finite implication for FDs and unary INDs, its cycle rules derive only unary FDs and unary INDs [1, Thm. 9.4.7]. Consequently, the FDs in Σ^+ are generated by the FDs of Σ and polynomially many unary FDs, all computable in polynomial time [23]. The first part of (4) inspects polynomially many unary INDs, and the second part is the test after the proof of Thm. 3.5, applied to these generators. □

The following example shows why condition (4) must be computed under finite implication.

Example A.8 (Finite implication matters). Let $R(K, X, A)$ have the primary key K, the unary IND $R[K] \subseteq R[X]$ and the declared FD $X \to A$, and let $Fz = \{K, X\}$. In every finite instance, X takes at least as many values as K, so X is a key, condition (4) holds, and every modification of A is order-insensitive. Under unrestricted implication, X is not a key, and the test would wrongly reject the schema.

For the class $\mathcal{A}$, the completion of a two-tuple core rests on the following chase.

Lemma A.9. *Let F be a set of FDs over a relation S, let K be a minimal key of S under F, and let V be a finite set of values for*

K. Some finite relation over S satisfies F and has exactly V as its projection on K.

Proof. For every value in V, create a tuple that takes this value on K and distinct nulls elsewhere. Chase with F: whenever two tuples agree on Z and $Z \to E \in F$, merge their values on E. For tuples u and v, let $M(u,v)$ be the attributes outside K on which they carry merged nulls, and let $A(u,v)$ be the attributes of K on which they agree. We show the invariant that $K \setminus \{B\} \to M(u,v)$ follows from F for all u, v and $B \in K$ with $u[B] \neq v[B]$. It holds initially, since M is empty. Let a step merge the E-values of p and q, which agree on Z with $Z \to E \in F$, and let it newly merge x and y, where x was merged with p and y with q on E before. If $x[B] \neq y[B]$, two neighbours in the sequence x, p, q, y differ on B. For the pairs x, p and q, y, the invariant already gives $K \setminus \{B\} \to E$. For the pair p, q, we have $Z \subseteq A(p,q) \cup M(p,q)$ and $A(p,q) \subseteq K \setminus \{B\}$, so the invariant gives $K \setminus \{B\} \to Z$ and therefore $K \setminus \{B\} \to E$. Now assume that the chase must merge two distinct values on some $B \in K$, because p and q agree on Z with $Z \to B \in F$ and $p[B] \neq q[B]$. The invariant gives $K \setminus \{B\} \to Z$, hence $K \setminus \{B\} \to B$, which contradicts the minimality of K. Hence the chase never merges values of K, and replacing every class of nulls by a new value yields the relation. □

Proof of Thm. 4.2. For a relation R, let F_R be its declared FDs.

(4)⇒(3). By Lem. A.5, applied to every F_R, every minimal key of every relation is frozen. The right side of every IND is such a key, so every IND belongs to $\mathcal{G}$. As in the proof of Thm. 3.5, every elementary FD of F_R^+ belongs to $\mathcal{G}$, since a key under F_R is a key under Σ. Consequently, $\mathcal{G} \models \Sigma$. Thm. 3.4 gives (3)⇒(2), and (2)⇒(1)⇒(5) is immediate.

(5)⇒(4). Assume that F_R violates condition (4) of Thm. 3.5. The proof of that theorem yields tuples t_1, t_2 and t_1', t_2' over R such that both pairs satisfy F_R and either intermediate state with one changed entity violates F_R. We complete both pairs with the same tuples W. Process the relations that the IND graph reaches from R in topological order. For such a relation S, collect the values that the tuples built so far, including t_1, t_2, t_1', t_2', carry on the left sides of the INDs into S. All these INDs reference the same minimal key of S, so Lem. A.9 builds the tuples of S. The process terminates because the IND graph is acyclic, and it adds no tuple to R. All other relations stay empty. Then $\{t_1, t_2\} \cup W$ and $\{t_1', t_2'\} \cup W$ satisfy Σ, contain the same entities, and agree on frozen values. This update has exactly two changed entities and is not executable. □

For the class $\mathcal{K}$, both directions work IND by IND.

Proof of Thm. 4.4. (4)⇒(3). The primary keys are frozen keys and belong to $\mathcal{G}$. Let $R[PD] \subseteq S[LC]$ satisfy (4). If $C \subseteq Fz$, its right side lies in $Fz \subseteq M^*$. Otherwise, $P \subseteq Fz$ and every D_i is frozen or copies a frozen C_i. In the latter case, the rule of Def. 3.1 puts D_i into $M^*(R)$, because L is frozen and P sits at its positions. The left side then lies in $M^*(R)$, the right side contains the frozen key L, and the IND belongs to $\mathcal{G}$. Hence $\mathcal{G} \models \Sigma$. Thm. 3.4 gives (3)⇒(2), and (2)⇒(1) is immediate.

(1)⇒(4). Let $\rho = R[PD] \subseteq S[LC]$ violate (4). Then $C \not\subseteq Fz$, and either some position i has a mutable C_i and a mutable D_i, or $P \not\subseteq Fz$ and every mutable C_i has a frozen D_i. Note that $R \neq S$, since the IND graph is acyclic. Let π, π_1, π_2, a and b be new constants. *Core of the first case.* An entity r of R points to π, and the entity s of S with key π has $s[C_i] = r[D_i] = a$ in I and b in J. All other copies and sources of ρ are 0. Either intermediate state with one changed entity leaves r without a witness. *Core of the second case.* Choose a mutable C_j, whose copy D_j is frozen. Two entities r_1 and r_2 of R differ on an attribute of $PK(R)$ outside P, which exists because P does not contain $PK(R)$. Their pointers π_1 and π_2 differ exactly on the mutable attributes of P, and $r_1[D_j] = a$ and $r_2[D_j] = b$. The entities s_1 and s_2 of S with keys π_1 and π_2 have $s_1[C_j] = a$ and $s_2[C_j] = b$. In J, r_1 points to π_2, r_2 points to π_1, $s_1[C_j] = b$ and $s_2[C_j] = a$. All other copies and sources of ρ are 0. Whichever of the four entities is written first, some entity of R loses its witness: a new r_1 or r_2 finds no matching tuple yet, and a new s_1 or s_2 strips the old r_1 or r_2. *Completion.* The constants π, π_1 and π_2 have pairwise distinct new components, and the primary keys of core tuples consist of new constants. Every other attribute of a core tuple that occurs in a pointer gets a new constant, and every remaining attribute gets 0. Every relation gets a base tuple of zeros. For every core or support tuple and every IND $Q[P'D'] \subseteq T[L'C']$ of its relation other than ρ, the relation T needs a witness whose key is the value of the tuple on P' in I or J. Unless a core tuple has this key, T gets a support tuple with this key and 0 elsewhere, and the process terminates because the IND graph is acyclic. Such a support tuple is a witness, since the copies of the referencing tuple are 0 and the sources of T lie outside its key. Core tuples lie in R and S, and support tuples never reach R. A tuple can need the key π, π_1 or π_2 of S only through an IND without copies. Indeed, the other pointers of R into S differ from P in some position, and a support tuple carries 0 outside its key, so the pointer of an IND with copies, which leaves the primary key, yields a key with a component 0. The core tuple with this key is then a witness, since there are no copies to match. Hence every relation keeps distinct keys, every core violation persists, and the update is not executable. □

Both restrictions on pointers in $\mathcal{K}$ are needed. Firstly, let the relation $R(A, B, D)$ have the primary key A, let $S(K_1, K_2, C)$ have the primary key K_1K_2, and let only B and C be mutable under $R[ABD] \subseteq S[K_1K_2C]$, whose pointer contains $PK(R)$. Condition (4) fails, but an entity of S can only be referenced by the entity of R whose key it carries. Hence writing first the entities of S that gain their referencing entity, then the entities of R, and last the other entities of S executes every modification update. Yet an update that moves an entity of R away from an entity of S and changes that entity is not order-insensitive. Secondly, let $T(K_1, K_2, V)$ be frozen with the primary key K_1K_2, let $R(K, A, B, D)$ have the primary key K and the frozen attributes B and D, and let $S(L, E)$ have only its primary key L frozen. The INDs are $R[AD] \subseteq S[LE]$, $R[AB] \subseteq T[K_1K_2]$ and $T[K_1V] \subseteq S[LE]$, and the pointer of the third lies inside $PK(T)$. Condition (4) fails for the first IND, since A and E are mutable. Yet the key L makes the three INDs imply $R[ABD] \subseteq T[K_1K_2V]$, whose right side is frozen, and this IND and the third one imply the first. Hence $\mathcal{G} \models \Sigma$, and the schema is order-insensitive by Thm. 3.4. As in Ex. A.12, the frozen relation T fixes the values that referenced entities of S carry.

Proof of Prop. 4.6. Let (I, J) be a fresh modification update, let $\rho = R[PD] \subseteq S[LC]$ be an IND of Σ, and let r be an entity of R. Let s_I and s_J be the entities of S whose keys are $I(r)[P]$ and $J(r)[P]$. They exist and are unique, since L is a frozen primary key, and every intermediate state needs one of them as the witness of r under ρ.

Assume that every position i has $C_i \in Fz$ or $D_i \in Fz$. Then s_J keeps its sources. Indeed, for a mutable C_i the copy D_i is frozen, so $J(s_J)[C_i] = J(r)[D_i] = I(r)[D_i]$ is a value of C_i in I, which a fresh update writes into no source. Hence the new tuple of r has its witness in every intermediate state, and the old tuple of r lacks it exactly when s_I is new and has changed a source of ρ. Since the primary keys are frozen and are the only FDs, an intermediate state $I \oplus T$ is legal exactly when T contains every such r whenever it contains s_I. Each of these requirements leads from an entity of R to an entity of S along an edge of the IND graph, so writing the relations in a topological order of the IND graph, referencing relations first, keeps every prefix state legal. If condition (4) of Thm. 4.4 holds as well, every modification update is order-insensitive by that theorem.

Conversely, let some position i have a mutable C_i and a mutable D_i. The core of the first case in the proof of Thm. 4.4, completed as there, changes only r and s and writes the new constant b, so it is a fresh update with two changed entities that is not executable. Finally, let every position satisfy the condition, and let condition (4) of Thm. 4.4 fail for ρ. Then $P \not\subseteq Fz$ and some C_j is mutable. An entity r of R points to π_1 in I and to π_2 in J, where π_1 and π_2 differ exactly on the mutable attributes of P, and the entities s_1 and s_2 of S with these keys have $s_1[C_j] = s_2[C_j] = r[D_j] = a$ in I. Besides r, only s_1 changes, to $s_1[C_j] = b$, and all other copies and sources of ρ are 0. After the same completion, the update is fresh, since b and π_2 are new, and writing s_1 before r strips the old r of its witness. □

A bounded SAT search confirms Prop. 4.6, including the order of the relations, on 17,436 pairs of 403 random schemas of class $\mathcal{K}$ and freezes, with up to five entities and five values, and finds a witness with two changed entities wherever a condition fails.

Proof of Thm. 4.7. Let M be the attributes that the template modifies. Its updates are the fresh modification updates under the freeze Fz_M of all attributes outside M, and it needs only order exactly when all of them are executable. If Σ contains only FDs, Prop. 3.7 under Fz_M shows that all of them are order-insensitive, and hence executable, if no elementary FD $X \to A$ in Σ^+ whose left side X is not a superkey has $A \in M$, and that otherwise one of them is not executable. If Σ belongs to $\mathcal{K}$, Prop. 4.6 under Fz_M shows that all of them are executable if and only if no IND $R[PD] \subseteq S[LC]$ in Σ has both $C_i \in M$ and $D_i \in M$ at some position i. □

The running example illustrates Thm. 4.2.

Example A.10 (Without the cached path). Without the cached column *gpath*, Ex. 1.1 keeps a foreign key to a primary key, so the schema belongs to $\mathcal{A}$. Thm. 4.2 then makes every modification order-insensitive, renames and moves included. It is the cached copy, not the foreign key, that defeats write order in Ex. A.2.

Unique keys. Unique keys besides the primary key can pin pointers. Indeed, if a referencing entity holds frozen copies of attributes that form a unique key of S, then it can never move to another entity of S, since that entity would have to carry the same values. Let Σ belong to $\mathcal{K}$, except that its relations may also declare unique keys, which no IND references. For a set M of attributes and an IND $\rho = R[PD] \subseteq S[LC]$, let $X_\rho(M)$ be the least set of attributes of S that contains every attribute of LC whose counterpart in PD lies in M, and the copies of every IND $S[P'D'] \subseteq T[L'C']$ with $P' \subseteq X_\rho(M)$. Let Fz^* be the least set $M \supseteq Fz$ that contains the pointer P of every IND ρ for which $X_\rho(M)$ contains a unique key of S. It exists, since $X_\rho(M)$ grows with M, and it results from adding such pointers to Fz until none is missing. The iteration matters when pointers overlap. Once Fz^* contains the pointer of ρ, the attributes that it shares with the pointer of another IND ρ' out of R enter M, and their counterparts may complete a unique key in $X_{\rho'}(M)$.

Proposition A.11 (Unique keys). *For such schemas, Prop. 4.6 holds with Fz^* in place of Fz.*

Proof. Let (I, J) be a fresh modification update under Fz. Firstly, every intermediate state satisfies the unique keys. Indeed, let u and v be distinct entities of a relation with a unique key U, and consider the old tuple of u next to the new tuple of v. If v keeps its values on U, the two tuples differ on U since I is legal, and otherwise v writes into U a value that is new in its column, which u does not carry in I. Secondly, (I, J) changes no attribute of Fz^*. We show by induction along the iteration that if (I, J) changes no attribute of a set $M \supseteq Fz$, then it changes no pointer that the next step adds. Indeed, let $X_\rho(M)$ contain a unique key U of S, let r be an entity of R, and let s_I and s_J be the entities of S whose keys are $I(r)[P]$ and $J(r)[P]$. Let A be an attribute of LC whose counterpart B in PD lies in M. Then $J(s_J)[A] = J(r)[B] = I(r)[B] = I(s_I)[A]$. If A lies in the key L, which is frozen, then $I(s_J)[A] = J(s_J)[A]$. If A is a source, then $J(s_J)[A]$ is a value of A in I, so the fresh update has not written it, and again $I(s_J)[A] = J(s_J)[A]$. The copies of an IND $S[P'D'] \subseteq T[L'C']$ are determined by P', since L' is the primary key of T. Hence s_I and s_J agree on $X_\rho(M)$ in I, so on U, and $s_I = s_J$, so r keeps its pointer P. The fresh modification updates under Fz and under Fz^* thus coincide, and the unique keys hold in every intermediate state. Then the first part of the proof of Prop. 4.6 applies with Fz^* in place of Fz. If condition (4) holds for Fz^* as well, then either s_I and s_J keep their sources, or $s_I = s_J$ and every position i has C_i or D_i unchanged, so every combination of old and new tuples of r, s_I and s_J satisfies ρ, and every intermediate state is legal.

Conversely, the counterexamples of the proof of Prop. 4.6, taken under Fz^*, are fresh modification updates under Fz, and they carry over once the completion gives every support tuple its own new constants outside its key and every copy the value of its source. Two tuples of one relation then agree on an attribute only where the construction forces them to. This happens only for the old and the new witness of an entity r that moves under some IND ρ whose pointer P is not contained in Fz^*. The two witnesses agree on the attributes of L whose counterparts in P lie in Fz^*, on the sources whose copies lie in Fz^*, and on the copies that these attributes determine, that is, on $X_\rho(Fz^*)$, and on nothing else. Since Fz^* is a fixpoint, $X_\rho(Fz^*)$ contains no unique key of S. Hence the two

witnesses differ on an attribute of every unique key, and every counterexample respects the unique keys. □

The bounded SAT search confirms Prop. A.11 on 4,172 pairs of random schemas and freezes with one or two unique keys of one or two non-key attributes, where six pairs need seven entities for their witness. Under Fz in place of Fz^*, 157 of these pairs would be predicted wrongly. Pointers that overlap need the iteration. On 8,498 pairs of 144 schemas in which two INDs out of one relation share pointer attributes, with unique keys on sources, on sources together with key attributes and on copies, the search finds a counterexample for every pair that Prop. A.11 rejects, with up to eight entities, and none for the others with up to five. Computing every set X_ρ under Fz alone, without the iteration, rejects 28 pairs wrongly.

Conditional INDs. Part (2) of Thm. 4.8 restricts the pointers in two ways: (a) every CIND with a condition on S has a pointer $P \not\subseteq PK(R)$, and (b) if such a CIND has no condition on R, it shares its pointer and S with no other CIND. Without these restrictions, a CIND can imply another, and the condition of part (2) remains sufficient but is no longer necessary. For (b), let $Project[gid] \subseteq Group[gid; arch \in \{0\}]$ join the CIND of Sec. 4.3, which it implies, and let $Project.gid$ be frozen while both flags are mutable. The condition fails for the CIND of Sec. 4.3, which tests the mutable flag of projects, yet every group with a project keeps $arch = 0$ in I and in J, so every modification update is order-insensitive. For (a), let $R[P] \subseteq T[K]$ and $T[K] \subseteq S[L; g \in B]$ with $K = PK(T)$ imply $\rho = R[P] \subseteq S[L; g \in B]$, and let P and g be mutable. The condition fails for ρ, yet an entity of S to which an entity of R points in I or in J shares its key with an entity of T in both instances, which needs it to meet $g \in B$ in both, so every modification update is order-insensitive. On CINDs, we call a modification update *fresh* if, beyond Sec. 3.3, every value that it writes into an attribute that a CIND tests on its right side is new in its column. The proof reads the precedence graph of Prop. A.1 through enabled and disabled entities.

Proof of Thm. 4.8. (1) The argument of Prop. A.1 applies to CINDs, since a condition only selects the tuples that need a witness and the tuples that can serve as one, and $L = PK(S)$ identifies one entity. Hence the update is binary. An edge from v to u either leaves a referenced entity v whose new tuple alone serves the new tuple of u, so that the update enables v, or enters a referenced entity u whose old tuple alone serves the old tuple of v, so that the update disables u. No update both enables and disables an entity e of S. Indeed, both tuples of e would be present, with the old value of g in $B_2 \setminus B_1$ and the new one in $B_1 \setminus B_2$ for the sets B_1 and B_2 of two CINDs into S, which form a chain. Every edge thus leaves an enabled entity or enters a disabled one, and it joins two relations that the acyclic IND graph orders strictly. Consider the order of part (1) and an edge from v to u. If the edge leaves an enabled v, then v lies in the first group, and u lies in a later group or is an enabled entity of a relation that references the relation of v, which the first group writes later. Symmetrically, if the edge enters a disabled u, then u lies in the last group, after every entity of the earlier groups and after the disabled entities of the relations that reference the relation of u. No edge joins two entities of the middle group. Hence the order respects every edge, every prefix is closed under predecessors, and every prefix state is legal. Part (1) does not use the restrictions of part (2).

(2) Let every CIND that tests a mutable g have a frozen pointer and test no mutable f. A modification keeps every entity present, so a CIND that tests a frozen g or no attribute of S yields no edge. For a CIND that tests a mutable g, a tuple of R that needs a witness in I needs it in J at the same entity, and conversely, so this entity meets the condition on S in both versions, and the CIND yields no edge either. Hence the precedence graph has no edge, and every intermediate state is legal. This direction does not use the restrictions either.

Conversely, let $\rho = R[P; f \in A] \subseteq S[L; g \in B]$ test a mutable g, and let $P \not\subseteq Fz$ or f be mutable. For every relation whose CINDs test an attribute on the right side, fix a value b_1 in the least set of its chain, and for S also a value b_0 outside every set of its chain. The update changes two entities, r of R and s of S, where s carries $g = b_0$ in I and $g = b_1$ in J, and both keep all other values. If $P \not\subseteq Fz$, then r meets $f \in A$, if ρ tests f, in I and J, and it points to s in J and to an unchanged entity s' with $g = b_1$ in I, whose key differs from that of s exactly at the mutable attributes of P. If $P \subseteq Fz$, then f is mutable, r points to s in I and J, and f changes from a constant outside every set that a CIND tests f against to a value in A. The keys of s and s' and every other attribute of r, s and s' that no CIND tests carry new constants, and every other tested attribute on the left carries a constant outside all sets of its CINDs. Top-down along the acyclic IND graph, the completion then gives every tuple that needs a witness in I or J an unchanged entity at the key that it points to, with b_1 in the tested attribute, if any, and new constants in its other attributes. Hence every pointer of a completion entity into S under a CIND with a condition on S contains an attribute outside the key by restriction (a), whose value differs from the new constants in the key of s, so it never points to s.

I and J are legal. Every completion entity and s' meet every condition on their right side, since b_1 lies in every set of their chains, and they exist in both instances. The new tuple of s meets every condition on S as well. In J, the new tuple of r needs at s at most these conditions, and in I it points to s' if $P \not\subseteq Fz$. If $P \subseteq Fz$, then the old tuple of r meets no condition on R, so it needs at s only the CINDs without a condition on R, which by restriction (b) test no attribute of S, since they share the pointer and S with ρ. Hence no tuple of I needs the old tuple of s to meet a condition on S, and every other demand of I and J is met by construction. The intermediate state that shows only r new is illegal, since the new tuple of r meets $f \in A$ and points to the old tuple of s, which fails $g \in B$. The update is thus not order-insensitive.

The reverse update (J, I), with b_0 new in the column of g and, if $P \subseteq Fz$, the constant of f in I new in its column, is fresh, since it writes no key and no other tested value. It disables s, and its intermediate state that shows only s new is illegal, so part (2) also holds for fresh updates. □

For modification updates, the proof uses only that L identifies one entity in every state. It therefore applies also when L is a unique key of S that no update modifies, as for the flags that Sentry sets for accounts provisioned by an identity provider (Sec. 6.2).

Beyond the exact classes. Outside the three classes, write order can suffice for INDs although the condition of Thm. 3.4 fails, as the following example shows for frozen reference data.

Example A.12 (Frozen reference data). Let $Ref(L, Q)$ be a reference table with primary key L that the workload never modifies. Let $R(N, P, D)$ have the primary key N, the mutable pointer P and the frozen copy D, and let $S(L, C)$ have the primary key L and the mutable source C. The INDs $R[P, D] \subseteq Ref[L, Q]$ and $R[P, D] \subseteq S[L, C]$ require every P-value to lead to a tuple of Ref and a tuple of S that both carry the D-value. Since L is a key of Ref and Ref never changes, the C-value that a referenced entity of S must carry is fixed by its key. Hence every modification update is executable: write first the entities of S that gain referencing entities, then the entities of R, and last the entities of S that lose them. Writing first an entity of S that loses its referencing entities strips them of their witness, so such updates are not order-insensitive. The schema is order-sufficient but not order-insensitive, and $\mathcal{G} \not\models \Sigma$.

Two INDs into one relation can force a cycle of four writes although each is harmless alone. Let $T(P, V)$ be a frozen reference table with key P, let $R(K, A, D)$ and $Q(L, B, G)$ have frozen keys and the frozen attributes D and G, and let $S(C, E, F)$ have the frozen key C. The INDs are $R[AD] \subseteq T[PV]$, $Q[BG] \subseteq T[PV]$, $R[AD] \subseteq S[CE]$ and $Q[BG] \subseteq S[CF]$. Each IND into S alone behaves as in Ex. A.12. Together, let T hold $(1, 0)$ and $(2, 0)$, let r move from 1 to 2 and q from 2 to 1, and let s_1 and s_2 with keys 1 and 2 change (E, F) from $(0, 1)$ to $(1, 0)$ and from $(1, 0)$ to $(0, 1)$. Then r waits for s_2, s_2 for q, q for s_1, and s_1 for r, so no order exists. We conjecture that the condition $\mathcal{G} \models \Sigma$ remains exact for acyclic INDs if no pointer contains the primary key of its own relation, no IND derives a key, and no IND transfers an FD from frozen reference data. The evidence for this conjecture compares the condition with the semantics. The semantics enumerates small modification updates, and a SAT search over larger updates covers the cases in which the condition fails but enumeration finds no cycle. A random generator combines composite pointers, pointers that meet the primary key, attributes shared by two INDs and chains of copies. It confirms the decider of class $\mathcal{K}$ on all 33,445 pairs inside the class, and the condition and the semantics agree on all 5,400 pairs in the class of the conjecture. Outside this class, the condition fails while write order suffices on 1,039 of 46,556 pairs, and each of them has a pointer that contains the primary key of its relation or ranges over frozen reference data. In all but 10 of these pairs some update is not order-insensitive. In the remaining ones a frozen copy fixes the attribute it copies in every referenced entity, so every update is order-insensitive although $\mathcal{G} \not\models \Sigma$.

A.4 Proofs for Sec. 5

The conflict graph rewrites condition (4) of Thm. 4.4 as a set of binary clauses.

Proof of Thm. 5.1. Freezing $Z \subseteq W$ leaves the attributes in $W \setminus Z$ mutable. For an IND $R[PD] \subseteq S[LC]$, condition (4) of Thm. 4.4 is equivalent to the conjunction of the clauses "p or c is frozen" for all $p \in P$ and $c \in C$, and "C_i or D_i is frozen" for all positions i. Indeed, if $C \subseteq Fz$, every clause holds. Otherwise, some $c \in C$ is mutable, the first clauses freeze P, and the second clauses are the rest of condition (4). A clause that mentions an attribute outside W holds, and the other clauses are the edges of H. Hence Thm. 4.4 makes Σ order-sufficient exactly when Z covers every edge of H.

Algorithm 2 Freeze

Require: schema Σ in $\mathcal{K}$, modified attributes W with weights w
Ensure: a freeze $Z \subseteq W$ that makes Σ order-sufficient, of minimum weight if H is bipartite and at most twice the minimum otherwise
1: $H \leftarrow$ the conflict graph of Σ on W, without the edges at pointers if only fresh updates must be executable
2: **if** H is bipartite with sides U and V **then**
3: link s to every $A \in U$ with capacity $w(A)$, every $B \in V$ to t with capacity $w(B)$, and A to B with capacity ∞ for every edge $\{A, B\}$ of H
4: $(S, T) \leftarrow$ a minimum s-t cut
5: **return** $(U \cap T) \cup (V \cap S)$
6: $r \leftarrow w$
7: **for all** edges $\{A, B\}$ of H **do**
8: $\varepsilon \leftarrow \min(r(A), r(B))$, $r(A) \leftarrow r(A) - \varepsilon$, $r(B) \leftarrow r(B) - \varepsilon$
9: **return** $\{A \in W : r(A) = 0\}$

For the hardness, let G be a graph whose edges are oriented along a numbering of its nodes, so that the orientation is acyclic. For every node u, the relation R_u has the primary key K_u, the attribute A_u, and an attribute D_{uv} for every edge from u to v. Every such edge contributes the IND $R_u[A_u D_{uv}] \subseteq R_v[K_v A_v]$. The schema belongs to $\mathcal{K}$: the pointer A_u is shared by the outgoing edges of u and is disjoint from K_u, every copy D_{uv} lies outside K_u and occurs in one IND only, and every source A_v lies outside K_v and is no copy. With $W = \{A_u\}$ and unit weights, H has the edge $\{A_u, A_v\}$ for every edge of G and no other edge, since the copies are frozen. MIN-FREEZE with bound k then asks whether G has a vertex cover of size at most k, which is NP-hard [39]. □

The tractable case of Sec. 5.1 follows from the same graph. Every edge of H joins attributes of two relations that an IND connects, so a 2-coloring of the undirected IND graph colors H. A minimum-weight vertex cover of a bipartite graph is a minimum cut of the standard flow network. The following example weighs the minimal freezes of the running example.

Example A.13 (Minimal freezes). In Ex. 4.5, H has an edge between *Project.gid* and *path* and one between *path* and *gpath*. Its minimal vertex covers are $\{path\}$ and $\{Project.gid, gpath\}$. Freezing *path* requires rewriting every rename, whereas freezing *gid* and *gpath* requires rewriting every move and every rename of a group with projects. Hence freezing *path* is cheaper unless renames of empty groups outnumber moves.

Alg. 2 computes the minimum cut on a bipartite conflict graph (lines 2–5) and the 2-approximation of Bar-Yehuda and Even otherwise (lines 6–9).

For shard keys, sufficiency reuses Lem. A.4, and necessity reuses the two-tuple constructions of Thm. 3.5.

Proof of Thm. 5.2. *Sufficiency.* Write the shards one at a time, each in one atomic step and in any order. Assume that two tuples t and u violate an elementary FD $X \to A$ of Σ^+ in some intermediate state. By Lem. A.4, the FD is unstable. The tuples of one shard are all old or all new, and I and J satisfy Σ, so t and u lie on different shards. They agree on $X \cap Fz$, whose values never change, so their tuples in I agree on $X \cap Fz$ and hence on $(X \cap Fz)^+ \supseteq SK$. Since

$SK \subseteq Fz$, the tuples t and u agree on SK and lie on the same shard, a contradiction.

Necessity. We first reduce the condition to two weaker ones. For every unstable elementary FD $X \to A$ in Σ^+, let (i) require $X^+ \supseteq SK$ if A is mutable, and let (ii) require $(X \setminus \{x\})^+ \supseteq SK$ for every mutable $x \in X$ if A is frozen. Assume (i) and (ii). We show $(X \cap Fz)^+ \supseteq SK$ by induction on $|X \setminus Fz|$, where an FD with a frozen right side precedes one with a mutable right side of the same size. Let $Y = X$ if A is mutable. Otherwise, A is frozen, X contains a mutable x because the FD is unstable, and we let $Y = X \setminus \{x\}$. Then $Y^+ \supseteq SK$ by (i) or (ii). Let $s \in SK$. If $s \in Y$, then $s \in X \cap Fz$ because $SK \subseteq Fz$. Otherwise, some $Y' \subseteq Y$ gives an elementary FD $Y' \to s$. If $Y' \subseteq Fz$, then $s \in (X \cap Fz)^+$. Otherwise, $Y' \to s$ is unstable, since a frozen key inside Y' would make $X \to A$ stable. It has a frozen right side and $|Y' \setminus Fz| \leq |Y \setminus Fz|$, so it precedes $X \to A$. The induction hypothesis gives $(Y' \cap Fz)^+ \supseteq SK$, and $s \in (X \cap Fz)^+$ because $Y' \cap Fz \subseteq X \cap Fz$.

It remains to refute (i) and (ii) by updates. If (i) fails, take two tuples whose agree set is $C = X^+$ and give both the same new value on A, as in Case 2 of the proof of Thm. 3.5. If (ii) fails, take two tuples whose agree set is $C = (X \setminus \{x\})^+$ and swap their mutable parts, as in Case 3. In both cases $SK \not\subseteq C$, so the two tuples differ on SK, and some placement puts them on different shards. Every execution by single-shard steps then passes through a state in which exactly one of them is new, and this state violates $X \to A$. □

The condition of Thm. 5.2 uses closure and not containment, as the following example shows.

Example A.14 (Closure, not containment). Let a relation (*Id*, *Zip*, *City*, *Tax*) have the key *Id* and the FDs *Zip* → *City* and *City* → *Tax*, and let only *Tax* be mutable. The shard key *City* meets the condition for the unstable FD *Zip* → *Tax*, since tuples that agree on *Zip* lie in the same city.

On the running example, the condition selects the shard key as follows.

Example A.15 (Sharding projects by group). Consider again Ex. 3.6, where *Project* stores the path under *gid* → *gpath*, and let the workload rename groups but never move projects. The FD *gid* → *gpath* is unstable, and the condition requires $\{gid\}^+ \supseteq SK(Project)$, so $SK(Project) = \{gid\}$, the only frozen attribute in $\{gid\}^+$. Sharding projects by their group turns every rename into one single-shard step, whereas sharding them by *pid* requires atomic visibility.

Under Thm. 5.3, the cached path of Ex. 4.5 is carried as well if groups form trees, every project copies the frozen root of its tree from its group, and projects move only within their tree: sharding both relations by the root then carries the cached-path IND.

For fresh updates, the proof of Prop. 3.7 shows that the two tuples of a violation agree on X already in I, and hence on X^+. Sufficiency then needs $X^+ \supseteq SK$ only for the elementary FDs that Prop. 3.7 retains, and necessity uses the construction for (i) alone.

Class $\mathcal{K}$ yields binary updates, and the proof of the next theorem reads the edges of $\mathcal{P}$ off the INDs.

Proof of Thm. 5.3. Let (I, J) be a modification update, let $\rho = R[PD] \subseteq S[LC]$ be an IND of Σ, and let r be a changed entity of R. As in the proof of Prop. 4.6, let s_I and s_J be the entities of S whose keys are $I(r)[P]$ and $J(r)[P]$. The old tuple of r has a witness under ρ exactly when s_I is old or $J(s_I)[C] = I(r)[D]$, and the new tuple has one exactly when s_J is new or $I(s_J)[C] = J(r)[D]$. An unchanged entity of R keeps its witness, since its referenced entity has the same key in I and J, and I and J are legal. The primary keys are frozen and are the only FDs, so the update is binary, and the graph whose edges lead from some r to s_I with $J(s_I)[C] \neq I(r)[D]$ and from some s_J to r with $I(s_J)[C] \neq J(r)[D]$ is a precedence graph.

Sufficiency. Let ρ satisfy condition (4). If $C \subseteq Fz$, every entity of S keeps its sources, and the two inequalities fail because I and J are legal. Otherwise $P \subseteq Fz$, so $s_I = s_J$, and at every position i the source C_i or the copy D_i is frozen. In both cases $I(s_I)[C_i] = I(r)[D_i] = J(r)[D_i] = J(s_I)[C_i]$, and again neither inequality holds. Hence every edge of $\mathcal{P}$ stems from an IND that violates condition (4) and therefore carries the shard key. Its relations share a group, and since I and J satisfy Σ, $I(r)[SK(R)] = I(s_I)[SK(S)]$ and $J(r)[SK(R)] = J(s_J)[SK(S)]$. Shard keys are frozen, so r, s_I and s_J agree on their shard-key values and lie on one shard under every placement. Every edge of $\mathcal{P}$ thus stays within one shard, so does every strongly connected component, and Lem. 5.5 (1) executes the update.

Necessity. Let ρ violate condition (4) and not carry the shard key. The update (I, J) of the proof of Thm. 4.4 for ρ has a cycle of $\mathcal{P}$ through an entity r of R and its witness s under ρ, namely $r \to s \to r$ in the first case and $r_1 \to s_1 \to r_2 \to s_2 \to r_1$ in the second. If R and S lie in different groups, some placement puts r and s on different shards. Otherwise Σ does not imply the equality of the shard keys, so some finite instance M of Σ has a tuple r' of R whose witness s' under ρ has $s'[SK(S)] \neq r'[SK(R)]$. FDs and INDs are preserved under direct products [29], so $I \times M$ and $J \times M$, with pairs of values encoded as constants, satisfy Σ, and they form a modification update, since M stays fixed. In the product, (r_k, r') references (s_l, s') under ρ in a state exactly when r_k references s_l in the corresponding state of (I, J), and the second components agree because s' is the witness of r'. Hence every edge between r_k and s_l lifts to one between (r_k, r') and (s_l, s'), and the cycle lifts to $(r, r') \to (s, s') \to (r, r')$, or to $(r_1, r') \to (s_1, s') \to (r_2, r') \to (s_2, s') \to (r_1, r')$. The shard-key values of (r, r') and (s, s') differ in the second component, so some placement puts them on different shards. In both cases a cycle of $\mathcal{P}$ spans two shards under some placement, and Lem. 5.5 (1) rules out every execution by single-shard steps. □

For fresh updates, the sufficiency part applies verbatim. The first case of the necessity part is a fresh update, also in the product, whose new source values are pairs with a new first component, so every IND with a mutable copy of a mutable source must carry the shard key. Neither bound is exact. An IND whose pointer and source are mutable and whose copy is frozen yields only edges from R to S under fresh updates, by the argument of the proof of Prop. 4.6, which applies to each IND separately. If it is the only IND that violates condition (4), every fresh update is executable entity by entity, so it need not carry the shard key. However, INDs can combine, for example two INDs between the same relations. Let $R(N, A, B, D, E)$ and $S(L, C)$ have the primary keys N and L, let only A, C and E be mutable, and let the INDs be $R[AD] \subseteq S[LC]$ and $R[BE] \subseteq S[LC]$,

with $SK(R) = B$ and $SK(S) = L$ in one group. The second IND carries the shard key, and the first has a frozen copy. Let r_1 and r_2 reference s_1 and s_2 under the second IND and s_2 and s_1 under the first, and let every value of C, D and E be 1. The fresh update that gives s_1 and s_2 new values of C, gives r_1 and r_2 the matching values of E, and moves both under the first IND to an unchanged entity whose C-value is 1 has the cycle $r_1 \to s_2 \to r_2 \to s_1 \to r_1$. Since r_i lies on the shard of s_i, the cycle spans two shards under every placement that separates the keys of s_1 and s_2, and the update is not executable under SK. Combinations through a third relation exist as well, so an exact condition has to follow paths of INDs. A bounded SAT search confirms Thm. 5.3 on 13,199 combinations of 280 random schemas of class $\mathcal{K}$, freezes, shard keys of one attribute and groups. It finds no counterexample with up to five entities and five values wherever the condition holds, and a counterexample with at most four changed entities wherever it fails.

Insertions and deletions need no precedence graph when the INDs are acyclic.

Proof of Prop. 5.4. Let the update only insert entities, so that $I \subseteq J$, and write the relations so that S precedes R whenever Σ has an IND from R to S. Every intermediate state L satisfies $I \subseteq L \subseteq J$, and a subset of J satisfies every FD that J satisfies. Let t be a tuple of L in a relation R, and let $R[X] \subseteq S[Y]$ be an IND. If t is old, its witness in I lies in L. If t is new, J holds a witness s of t, and s is old or belongs to S, whose new tuples are all written before those of R, since $S \neq R$ by acyclicity. For deletions, the order passes through the states of the insertion from J to I in the reverse order, which writes the relations in reverse topological order. □

The block lemma reads the precedence graph as the complete list of violations.

Proof of Lem. 5.5. Since the update is binary, an intermediate state $I \oplus T$ is legal if and only if no edge of $\mathcal{P}$ leads from an entity outside T to an entity in T. (1) Order the strongly connected components of $\mathcal{P}$ topologically and write each in one step, which touches one shard. After every step, the written entities are closed under predecessors, so the state after every step is legal. Conversely, consider the first step that writes an entity of a component that spans two shards. This step writes only part of the component, so some edge of the component leads from an unwritten entity to a written one, and the state after the step is illegal. (2) If every shard commits exactly once, an execution is an order of the shards. It is legal if and only if every edge between two shards leads from an earlier shard to a later one. Such an order exists if and only if the graph obtained by merging the entities of every shard is acyclic. □

Rounds refine model (1) by letting the transactions of one round commit in any order.

Proof of Prop. 5.7. A state is legal if and only if no edge of $\mathcal{P}$ leads from an unwritten entity to a written one, since the update is binary. Let an execution in d rounds be legal, and let $r(e)$ be the round of the transaction that writes the entity e. Consider an edge from v to u. If $r(v) > r(u)$, the state after round $r(u)$ writes u but not v. If $r(v) = r(u)$ and u and v lie in different transactions, the transaction of u may commit first. Both states are illegal, so $r(v) \leq r(u)$, with equality only if u and v share a transaction and hence a shard. Along a path that ends in e, the round therefore grows by at least one at every edge between two shards, and $d \geq r(e) \geq 1 + \lambda(e)$.

Conversely, every cycle of $\mathcal{P}$ lies within a strongly connected component and hence within one shard. Removing a cycle from a walk therefore keeps all its edges between two shards, so $\lambda(e)$ is also the largest number of such edges on a walk that ends in e. For an edge from v to u, a walk that ends in v extends to u, so $\lambda(v) \leq \lambda(u)$, and $\lambda(v) < \lambda(u)$ if the edge joins two shards. Hence v lies in the transaction of u or in an earlier round. After the transactions of the earlier rounds and any of the current round have committed, the written entities are therefore closed under predecessors, and every state is legal. □

A relay that validates what the update read faces the same constraints as the original rounds, and its locks protect it only from updates that respect them. Here an entity *e shares a constraint* with a tuple t of another entity if one of the two tuples, t or the tuple of e, agrees with the other on the left side of an FD or is a witness of the other under an IND. An entity shares a constraint with an entity u of an update if it shares one with the current, the old or the new tuple of u, and a relay reads an entity that is absent as absent. We call an update *fenced* if it changes an entity only under an exclusive lock and commits a new tuple only while it holds a lock on every other entity that shares a constraint with the tuple, taken before it read the entity. Under primary keys and INDs into them, these entities are the parents and the children of the tuple, and a witness is the entity that carries the referenced key, so a fenced update locks the parent of every child that it inserts or moves.

Proposition A.16 (Recovery). *Let Σ consist of primary keys and INDs into them, let the other updates be fenced and lead from legal states to legal states, and let a crash stop leveled commits after some round in a legal state. Let a relay commit the remaining rounds as a fenced update that, before round k commits, locks the entities of round k, then reads every entity that shares a constraint with one of them, and aborts unless all are as round $k-1$ left them. Then every state after a commit is legal, and the relay completes the update unless another update has changed one of these entities.*

An aborted relay leaves the update stopped after some round in a legal state, and its outbox tells the application which writes remain, so that it can plan them again on the current state or report the update as failed.

Proof of Prop. A.16. The crash leaves a legal state after some round i, which Prop. 5.7 ensures when no other update intervenes, and other updates lead from legal states to legal states, so it suffices to show that every commit of the relay does. If the relay aborts, none of its writes becomes visible. Otherwise, consider the commit of a round $k > i$, and let u be an entity of round k. The relay locks u exclusively before it reads the entities that share a constraint with u, and, being fenced, it holds a lock on the parents of the new tuple of u until round k has committed. Hence no other update changes these parents between the read and the commit. No other update commits a tuple of a child of u in this time either, since being fenced it would need a lock on u. A child that another update has moved away from u before the read was read as such, and one that it moves away after the read no longer shares a constraint with u. Hence,

at the commit, every entity that shares a constraint with an entity of round k is as the relay read it, which is as round $k-1$ left it. An IND into a primary key is violated by a tuple and the entity that carries the referenced key, so every violation after the commit either involves no entity of round k, and then it exists before the commit, or involves an entity of round k and one that is as round $k-1$ left it. In the second case the violation also exists in the state after round k of the original execution, which is legal by Prop. 5.7. Hence the state after the commit is legal. The relay overwrites only entities that it has locked and found as the crash left them, so it overwrites no change of another update. □

Deadlocks need waits for locks, which the protocol of Sec. 5.3 excludes.

Lemma A.17 (No hidden deadlocks). *If every lock request of leveled commits, for a write, a fence or a validation, fails instead of waiting, and an update whose request fails aborts its open transactions, then no update waits for a lock while it holds one, and no deadlock arises.*

Proof. A transaction of leveled commits acquires locks only while it executes its statements, never waits for a lock, and releases its locks when it commits or aborts. Afterwards, an update waits only for the commits of its own earlier rounds, and a commit acquires no lock. A deadlock is a cycle of updates in which each waits for a lock that the next one holds, so none arises. □

Retries with randomized backoff keep updates that fail on the same lock from failing again in step, but they guarantee no progress: under heavy contention most attempts fail (Sec. 6.3), and a relay whose locks fail stops the update as an abort does.

A pause between rounds yields such a state for every fast reader, since it orders the commits of different rounds in time.

Proof of Prop. 5.8. Times are real times, and the clocks of the writer and the reader measure durations alike. For a round i, let a_i be the time at which the last transaction of round i acknowledged its commit. Every transaction of round i commits before a_i, and every transaction of a later round k commits after $a_{k-1} + \delta \geq a_i + \delta$, since its commit is sent no earlier. The reader reads every shard s at one time t_s and sees on s exactly the transactions of s that committed before t_s (assumption 1), and $|t_s - t_{s'}| < \delta$ for all shards s and s'. If the reader sees no transaction of the update, it sees the initial instance, which is legal. Otherwise, let j be the latest round of which it sees a transaction, on a shard s'. If $j > 1$, then $t_{s'} > a_{j-1} + \delta$, and every transaction of a round $i < j$, on a shard s, committed before $a_i \leq a_{j-1} < t_{s'} - \delta < t_s$. Hence the reader sees every transaction of the rounds before j, some of round j and none of the later rounds. By the proof of Prop. 5.7, the entities that these transactions write are closed under predecessors, so the state that the reader sees is legal. □

Replicas shift the times at which a reader sees a shard by at most their lag.

Corollary A.18 (Replicas). *Let every shard be read from a replica that applies the commits of the shard in their order, at most Δ after they happen. Then Prop. 5.8 holds for every reader that reads each replica once, and all replicas within less than $\delta - \Delta$.*

Proof. A replica read at time t shows a prefix of the commits of its shard that contains every commit before $t - \Delta$ and none after t. Hence it shows exactly the transactions of its shard that committed before some time t' with $t - \Delta \leq t' \leq t$. Read times within less than $\delta - \Delta$ of each other thus yield such times t' within less than δ of each other, and the proof of Prop. 5.8 applies to them. □

The reading of the precedence graph in the proof of Prop. 5.7 also proves that concurrent updates need only merge into a legal instance.

Proposition A.19 (Concurrency). *Let updates that change pairwise disjoint sets of entities start from a common instance, and let their combination, whose changed entities are the union of theirs, be binary. If applying all of them yields a legal instance, then every interleaving of legal orders of the updates passes only through legal instances.*

Proof of Prop. A.19. Applying all updates yields a legal instance, so their combination is an update, and let $\mathcal{P}$ be its precedence graph. Since the combination is binary, an intermediate state of the combination is legal if and only if no edge of $\mathcal{P}$ leads from an unwritten entity to a written one. Applying a single update alone yields its final instance, which is legal and is the intermediate state of the combination that writes exactly the changed entities of this update. Hence no edge of $\mathcal{P}$ enters the changed entities of an update from those of another. A state of an interleaving writes a prefix of the order of every update. An edge into a written entity u therefore starts at an entity of the same update, and since the prefix states of that update alone are legal, it starts within the written prefix. □

In the example of Sec. 5.3, leveled commits hold the lock on g_2 in their open transactions: the deleting update locks g_2 in the transaction of its last round before it reads the projects of g_2, and the inserting update locks g_2 for sharing in a transaction that commits after the project.

B FURTHER EXPERIMENTAL RESULTS

This appendix reports the results of Sec. 6 that the argument of the body does not need.

Coverage of the exact classes. The exact classes of Sec. 4 cover most constraints of the 97 templates of Sec. 6.2. Foreign keys alone, which class $\mathcal{A}$ covers, constrain 25 of the 65 templates of Sentry and 22 of the 32 of GitLab. Another 16 of Sentry add copies of referenced columns over one hop, which class $\mathcal{K}$ covers together with the unique keys of Prop. A.11. Five templates are unconstrained, and the others maintain conditional INDs, derived values or copies over two hops.

An eager first round. Aborting a failed update before any write is visible costs one round trip. On two or three rows, ordered commits of one row each commit every row in the message that executes it and take up to a quarter less time with one client, but a row that fails leaves the earlier rows visible. Where no write can fail logically, an eager first round saves this round trip and makes leveled commits the fastest way of writing with one client at every size (Fig. 4).

Routing and placement. Routed through the coordinator of Citus, leveled commits still take about half the time of 2PC at 128 rows with one client, but not at 8 rows behind added delay. With 64 clients, the coordinator becomes the bottleneck. Both advantages of Sec. 6.3 persist on both engines with every node on cores of its own and 0.5 ms added to every path between nodes (at most 0.6 of the time, at least twice the throughput).

Ordered 2PC.. Atomicity on top of order costs more, though less than the 2PC of the engines. An ordered 2PC, which prepares all transactions of leveled commits before the first round commits (Sec. 5.3), takes 6 to 26% longer with one client and up to halves their throughput with 64 clients, yet it stays faster than the fastest 2PC. Indeed, each prepare flushes a log, for which clients contend. Leveled commits thus trade atomicity for this margin.

Recovery. Completing an interrupted update without the client costs little. An outbox, one row in every transaction of the first round, raises the latency of leveled commits by at most a fifth and leaves them faster than 2PC at every point of Fig. 4. When a proxy dropped a commit message to one Citus worker every half second, one 2PC reported success for every update, but 37 projects stayed orphans until the periodic recovery of Citus completed the lost commits, up to 38 s later at its default interval of one minute. A shorter interval narrows this window but cannot close it. Leveled commits with an outbox left no orphan, since a lost message stops them between rounds, where the state is legal, and the client or the relay then completes the update. Validation matters once another update intervenes. In 3,000 recoveries that raced with a conflicting update, a relay that re-applied the new tuples left 653 orphans and lost 381 updates. The relay of Prop. A.16 left none, completing two thirds of the updates and stopping the others in a legal state.

Fences and deadlocks. Concurrent updates that do not merge into a legal instance need the locks of Sec. 5.3. When clients added projects to groups that other clients merged, 24% of the merges left an orphan under leveled commits and 68% under one 2PC. With the locks, no merge did, merges took as long as before, and additions took 0.85 instead of 0.71 ms. Lock requests that fail instead of waiting keep these locks free of hidden deadlocks, at the price of retries. When 64 clients moved projects between hot groups, writes that waited closed cycles of waits across workers, which only a lock timeout broke, whereas writes that failed closed none and sustained 6,440 updates per second. One 2PC that waited for the same locks sustained at most 35, since only the distributed deadlock detection of Citus broke its cycles. With a lock timeout or no-wait locks it would fail fast like leveled commits, a configuration that we did not run. Yet 63.5% of their attempts failed, and their 99th percentile latency reached 184 ms. Locking the rows in one global order avoids the cycles as well and sustained 3,871 updates per second, and with 8 or 16 clients under milder skew, waits bounded by a timeout of 100 ms were faster than failing writes, so failing writes pay off only under heavy contention.

Decision time. Bounded search confirms the verdict of every decider on all 106,270 random pairs of schema and freeze that we tried. On conflict graphs with odd cycles, the 2-approximation of Alg. 2 stays within 1% of the linear-programming lower bound.

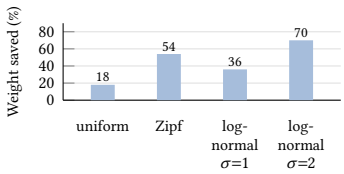


Figure 9: Weight Saved by the Optimal Freeze over Freezing Every Copied Source, by Frequency Distribution

Optimal freezing. Optimal freezing pays off once templates modify attributes at different frequencies. On the 179 shard-key copies of GitLab that class $\mathcal{K}$ covers, the conflict graph consists of stars around the sources, so under unit weights freezing every copied source is optimal. Fig. 9 shows that under synthetic frequencies, which we use since GitLab does not record them, the optimal freeze saves the more weight over this rule the more the frequencies are skewed. It saves a median 18% under uniform frequencies and 36 to 70% under log-normal ones. Indeed, a star whose copies weigh less than its source is covered by them, and skew makes such stars common.

Unique keys under sharding. A shard key that makes updates executable may still split a unique key, which a shard then cannot check. In GitLab, 1,040 of the 1,127 tables have a single-column shard key that makes every fresh update executable and that every unique key of the table determines through copies, so that a shard checks the key from local data, as Gupta and Widom [37] verify global constraints. The rule of Citus [21], which requires the shard key inside the primary key and every unique key, admits a single-column shard key for only 527 tables. The other 87 tables hold globally unique names, paths and tokens, which need a check across shards under any placement.